

ESTIMATING COEFFICIENT-BY-COEFFICIENT BREAKS IN PANEL DATA MODELS

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Abstract

When estimating structural breaks, existing econometric methods typically adopt an approach in which either all parameters change simultaneously or none of them do. In this paper, we consider the estimation of panel data models when an unknown subset of coefficients is subject to breaks. The challenge lies in estimating the breaks for each coefficient. To tackle this, we propose a new estimator for panel data, the “Coefficient-by-Coefficient Lasso” break estimator. This estimator is derived by penalizing the coefficients with a fused penalty and using component-wise adaptive weights. We present this estimator for two scenarios: those with homogeneous breaks and those with heterogeneous breaks. We show that the method identifies the number and dates of breaks for all coefficients with high probability and that the post-selection estimator is asymptotically normal. We examine the small-sample properties of the method through a Monte Carlo study and further apply it to analyze the influence of socioeconomic factors on crime.

Keywords: Breaks; Fused Lasso; Partial structural change; Panel data;

JEL Classification: C13; C23; C33; K42.

1 Introduction

Identifying structural breaks is recognized as a crucial task for economists and empirical researchers, especially in response to economic events or policy changes. As a result, there has been a surge of papers that suggest methods to infer structural breaks (see Qian and Su [2016](#), for an overview). The literature concerning structural breaks in time series is extensive, with

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CUSUM-based methods being central in the literature (see Brown et al. 1975). In contrast, the literature on structural breaks in panel data, where N cross-sectional units are observed over T periods, is much smaller. Most papers assume that there is only one break, whereas the literature addressing multiple structural breaks is more recent. A non-exhaustive list of papers that examine homogeneous multiple structural breaks in panel data includes Qian and Su (2016), Li et al. (2016), Boldea et al. (2020), and Kaddoura and Westerlund (2023). Moreover, heterogeneous structural breaks, defined as breaks that vary across different latent groups, have recently received some attention, as seen by Su et al. (2019), Okui and Wang (2021), and Lumsdaine et al. (2023). All these papers, whether focusing on single or multiple structural breaks, share a common characteristic: they estimate the break in the *slope vector* without providing information on which parameters are breaking.

In this paper, the term “vector breaks” refers to the conventional notion of structural breaks wherein the parameter vector shifts across different regimes. We also introduce the concept of “component-wise” or “coefficient-by-coefficient (CBC) breaks,” by which we mean that each component of the slope vector has its own set of breaks. Unlike vector breaks, where we estimate a single break number, m , along with a single set of break dates, $\{T_1, \dots, T_m\}$, the idea of CBC breaks is more general. One can still assume that the vector jumps if one or more coefficients jump, but the reverse is not necessarily true. Of course, in practice, a vector break may be observed if some of its components are breaking. However, it can be challenging to pinpoint which components caused the break, especially when many controls are included. Hence, knowing which components are shifting eases the interpretation of the results, which can be crucial in many applications.

In addition, allowing for vector breaks when breaks are not common leads to potentially overestimating the number of breaks for some or all of the parameters. This can cause a loss of statistical power, affecting forecasting, prediction, and the significance levels of tests. On the other hand, by allowing each coefficient to have its own set of breaks, we avoid this unnecessary loss of power. This highlights the importance of component-wise breaks: *efficiency and interpretability*.

Looking into recently published articles concerning structural changes shows that there is little to no discussion on why the analysis of breaks is focused on shifts in the slope vector. Qian and Su (2016) and Okui and Wang (2021) suggest shrinkage procedures that can infer the number of breaks and locations in the model. In addition, Boldea et al. (2020) suggest

using simple Least Squares and an Information Criterion to estimate the breaks. Other approaches include CUSUM and MOSUM statistics, as discussed by Hušková and Slabý (2001), Horváth and Hušková (2012), and Antoch et al. (2019). The closest attempts to address CBC breaks are those that discuss partial structural change models allowing shifts in a *predetermined* subset of the parameters (see, e.g., Bai and Perron 1998). Similar to our estimator, this is beneficial for obtaining more accurate estimates. However, this requires knowledge of the specific coefficients that are breaking, which is generally unknown. Even if one has such information, the other parameters are still assumed to have common breaks.¹ Having said that, these methodologies are limited to testing whether a vector of parameters is shifting, while the interest may lie in component-wise breaks.²

The approach in the present paper can be seen as a reaction to the previous discussions where we contribute to the literature by introducing novel approaches to estimate component-wise breaks. More specifically, we suggest an approach that penalizes each coefficient with an ℓ_1 fused penalty. This penalty induces sparsity of the component-wise breaks and enables us to use coefficient-specific weights which are important for the consistency of the estimators. Hence, viewing the breaks component-wise, using the ℓ_1 fused penalty, and having a component-wise weighting scheme is what distinguishes our approach to what is currently suggested by previous papers such as Qian and Su (2016) and Kaddoura and Westerlund (2023).³ Moreover, we consider two versions of the estimator to accommodate both homogeneous and heterogeneous breaks. The heterogeneous breaks estimator is defined for breaks that vary across different unknown groups. We show that the procedure for homogeneous (heterogeneous) breaks gives consistent estimates of the breaks and break dates as $N \rightarrow \infty$ ($N, T \rightarrow \infty$) for all coefficients and that the estimator of the coefficients is consistent and asymptotically normal. Both estimators, however, rely on N being large.

In the empirical illustration, we look into the relationship between crime and socioeconomic factors. We show how the method in this paper overcomes the issues discussed above giving us more reasonable results. Specifically, the effects of deterrence variables on crime

¹In papers such as Bai and Perron (1998), they still assume the notion of vector breaks even though they allow for some non-shifting parameters. Essentially, all they are doing is estimating vector breaks while keeping other parameters fixed. This leads to the same issues that are discussed in this paper.

²There is also a wide literature on maximum likelihood estimators on change-point models, as seen, for example, in Horváth (1993), Gombay and Horváth (1994), and Gombay and Horváth (1996).

³The approach suggested in this paper could also be utilized in a time-series setting ($N = 1$), but with the limitation of possibly over-estimating the component-wise breaks and requiring the regimes to have a large set of periods.

do not break; rather, it is the coefficients associated with the three that controls break, causing all other parameters to shift. This leads the estimates and significance levels to change emphasizing the sensitivity of this issue in practice. Moreover, in the Monte Carlo section, we showcase that the method performs well when used with small samples.

The rest of the paper is organized as follows: In section 2, we present the procedures for both homogeneous and heterogeneous cases. In Section 3, the asymptotics are presented. Section 4 provides Monte Carlo simulation examples to show the good small sample properties of the procedure in this paper. Section 5 focuses on an empirical illustration. The paper concludes in section 6. Post-selection definitions are provided in Appendix B, while technical proofs are in Appendices C and D. Algorithms can be found in Appendix E.

2 Estimation procedure

2.1 Homogeneous case: penalized estimator

Consider a scalar panel variable y_{it} , observed across $t = 1, \dots, T$ time periods and $i = 1, \dots, N$ cross-section units. Consider the following data-generating process (DGP):

$$y_{it} = \zeta_i + \mathbf{x}_{it}'\boldsymbol{\beta}_t + \varepsilon_{it}, \quad (2.1)$$

where $\mathbf{x}_{it} := [x_{1,it}, \dots, x_{p,it}]'$ is a $p \times 1$ vector of known regressors, $\boldsymbol{\beta}_t := [\beta_{1,t}, \dots, \beta_{p,t}]'$ is a conformable vector of unknown slope coefficients that can vary over time, ζ_i represents cross-sectional fixed effects that may be correlated with the regressors, and ε_{it} is a zero-mean error term. Throughout this paper, " $a := b$ " indicates that a is defined by b .

Let us begin by assuming that the k th component of $\boldsymbol{\beta}_t$, denoted as $\beta_{k,t}$, takes on $m_k + 1$ distinct numbers $\alpha_{k,j}$ such that

$$\beta_{k,t} = \alpha_{k,j} \quad (2.2)$$

for $t \in \mathbf{r}_{k,j} := \{T_{k,j-1}, \dots, T_{k,j} - 1\}$, $j = 1, \dots, m_k + 1$, $m_k \in \{0, \dots, T - 1\}$, $T_{k,0} = 1$, and $T_{k,m_k+1} = T + 1$ for all $k \in \mathcal{P} := \{1, \dots, p\}$. Hence, $\beta_{k,t}$ exhibits $m_k + 1$ distinct regimes, or m_k breaks, which occur at periods $t \in \mathfrak{T}_{m_k} := \{T_{k,1}, \dots, T_{k,m_k}\}$. At one end, we may have that $m_k = 0$ implying that there is only one regime with $\beta_{k,1} = \dots = \beta_{k,T} = \alpha_{k,1}$. At the other end, $m_k = T - 1$ means that we have $\beta_{k,t} = \alpha_{k,t}$ for all $t = 1, \dots, T$. These break dates allow us to derive $\mathbf{r}_k := \{\mathbf{r}_{k,j}\}_{j=1}^{m_k+1}$, a collection of regimes, where $\mathbf{r}_{k,j}$ is a "regime set" of dates within regime j for coefficient k . These regime sets will prove useful later on.

We aim to infer $\{\alpha_{k,j}\}$ and $\mathbb{T} := \{\mathfrak{T}_{m_k}\}_{k=1}^p$. It is then useful to define $\alpha_{k,j}^0, \beta_{k,t}^0$, and m_k^0 as the true values of $\alpha_{k,j}, \beta_{k,t}$, and m_k , respectively. Furthermore, we define $\mathbb{T}^0 := \{\mathfrak{T}_{m_k^0}\}_{k=1}^p$ as the collection of true break date subsets with $\mathfrak{T}_{m_k^0}^0 := \{T_{k,1}^0, \dots, T_{k,m_k^0}^0\}$ representing the set of true break dates. To estimate these parameters, we propose minimizing the following objective function to obtain the CBC Lasso (CBCL) estimates:

$$\mathcal{L}_{\lambda_1}(\{\beta_{k,t}\}) := \frac{1}{N} \sum_{i=1}^N \sum_{t=2}^T (\tilde{y}_{it} - \mathbf{x}'_{it}\boldsymbol{\beta}_t + \mathbf{x}'_{i1}\boldsymbol{\beta}_1)^2 + \lambda_1 \sum_{t=2}^T \boldsymbol{\omega}'_t \boldsymbol{\Delta}_t, \quad (2.3)$$

where we use deviations from the first time points $\tilde{y}_{it} := y_{it} - y_{i1}$ and include $\mathbf{x}_{i1}$ to eliminate the fixed effects,⁴ λ_1 is a tuning parameter, and $\boldsymbol{\Delta}_t := [|\beta_{1,t} - \beta_{1,t-1}|, \dots, |\beta_{p,t} - \beta_{p,t-1}|]'$ and $\boldsymbol{\omega}_t := [\omega_{1,t}, \dots, \omega_{p,t}]'$ are $p \times 1$ vectors of ℓ_1 fused terms and weights, respectively. Here, $\omega_{k,t}$ is a weight given by $\omega_{k,t} := |\dot{\beta}_{k,t} - \dot{\beta}_{k,t-1}|^{-\kappa_1}$ with $\kappa_1 > 0$ being a positive constant and

$$\dot{\beta}_{k,t} := \left(\mathbf{X}'_{k,t} \mathbf{M}_{\check{\mathbf{X}}_{-k,t}} \mathbf{X}_{k,t} \right)^{-1} \mathbf{X}'_{k,t} \mathbf{M}_{\check{\mathbf{X}}_{-k,t}} \tilde{\mathbf{y}}_t, \quad \text{for } t = 2, \dots, T \text{ and } k = 1, \dots, p, \quad (2.4)$$

where $\mathbf{X}_{k,t} := [x_{k,1t}, \dots, x_{k,Nt}]'$ and $\tilde{\mathbf{y}}_t := [\tilde{y}_{1t}, \dots, \tilde{y}_{Nt}]'$ are the $N \times 1$ variables stacked for each time point. The matrix $\mathbf{M}_{\mathbf{A}}$ is defined as the projection $\mathbf{M}_{\mathbf{A}} := \mathbf{I} - \mathbf{A}(\mathbf{A}'\mathbf{A})^{-1}\mathbf{A}'$, where $\mathbf{A}$ is a matrix with r rows and $\mathbf{I}$ is an $r \times r$ identity matrix. Hence, $\mathbf{M}_{\check{\mathbf{X}}_{-k,t}}$ represents a matrix that projects out $\check{\mathbf{X}}_{-k,t}$, where $\check{\mathbf{X}}_{-k,t} := [\mathbf{X}_{1,t} \cdots \mathbf{X}_{k-1,t} \quad \mathbf{X}_{k+1,t} \cdots \mathbf{X}_{p,t} \quad \mathbf{X}_{1,1} \cdots \mathbf{X}_{p,1}]$, an $N \times (2p-1)$ matrix of the $\ell \neq k$ regressors at time t , augmented with the first time points.

In (2.3), we deviate from the strand of literature by applying an ℓ_1 fused penalty with component-wise adaptive weights, as opposed to the commonly used grouped Frobenius fused penalty that induces vector breaks (see Qian and Su 2016; and Kaddoura and Westerlund 2023). There are two key reasons why this approach works. First, by penalizing the difference of each coefficient using the ℓ_1 norm, we induce component-wise breaks, allowing each coefficient to have its own set of breaks. Second, since we sum the ℓ_1 differences of the coefficients, we can incorporate coefficient-specific weights, $\omega_{k,t}$. This can be seen as a CBC weighting scheme, which is crucial for achieving consistency (see also Zou 2006).

Minimizing (2.3) yields the CBCL estimates, $\{\hat{\beta}_{k,t}\}$ and the estimated break dates, $\hat{\mathbb{T}} := \{\{\hat{T}_{1,1}, \dots, \hat{T}_{1,\hat{m}_1}\}, \dots, \{\hat{T}_{p,1}, \dots, \hat{T}_{p,\hat{m}_p}\}\}$, where $\hat{T}_{k,1} < \dots < \hat{T}_{k,\hat{m}_k}$ when $\hat{m}_k > 0$ for all k . We define the break dates of coefficient k when $|\hat{\beta}_{k,t} - \hat{\beta}_{k,t-1}| > 0$ for $t = \hat{T}_{k,1}, \dots, \hat{T}_{k,\hat{m}_k}$. If $|\hat{\beta}_{k,t} - \hat{\beta}_{k,t-1}| = 0$ for some k and for all $t = 1, \dots, T$, then we have $\hat{m}_k = 0$ and $\hat{\mathfrak{T}}_{\hat{m}_k} =$

⁴By including $\mathbf{x}_{i1}$ as additional regressors, the dimension of regressors increases by two. One could also use a time-demeaned transformation as done by Wang et al. (2018) for example. We opt away from this because it implies that we would need to include $\mathbf{x}_{i1}, \dots, \mathbf{x}_{iT}$ as additional regressors increasing the dimensions to $N \times Tp$.

$\widehat{\mathfrak{T}}_0 = \emptyset$. The estimated dates $\{\widehat{\mathfrak{T}}_{\widehat{m}_k}\}$, by definition, yield us with the regime estimates $\widehat{\mathfrak{r}}_{k,j} := \{\widehat{T}_{k,j-1}, \dots, \widehat{T}_{k,j} - 1\}$ of the true regime sets $\mathfrak{r}_{k,j}^0 := \{T_{k,j-1}^0, \dots, T_{k,j}^0 - 1\}$.

Let $\alpha^0(\mathbb{T}^0) := [\alpha_1^{0'}(\mathfrak{T}_{m_1}^0), \dots, \alpha_p^{0'}(\mathfrak{T}_{m_p}^0)]'$ be the true value of the post-selection vector $\alpha(\mathbb{T}) := [\alpha_1'(\mathfrak{T}_{m_1}), \dots, \alpha_p'(\mathfrak{T}_{m_p})]'$, where $\alpha_k(\mathfrak{T}_{m_k}) := [\alpha_{k,1}, \dots, \alpha_{k,m_k+1}]'$. To estimate the post-selection coefficients $\{\alpha_{k,j}^0\}$, we introduce “sub-regimes” to account for cases when coefficients $\ell \neq k$ may vary within $\mathfrak{r}_{k,j}$. For any $\ell \neq k$, consider the sequence of sets $\mathfrak{r}_{\ell,1} \cap \mathfrak{r}_{k,j}, \dots, \mathfrak{r}_{\ell,m_\ell+1} \cap \mathfrak{r}_{k,j}$. Let $\mathfrak{r}_{\ell,1}^{(k,j)}$ be the first instance of a non-empty set in the sequence, $\mathfrak{r}_{\ell,s_\ell}^{(k,j)}$ the last, and $\mathfrak{r}_{\ell,c}^{(k,j)}$ for $c \in \{2, \dots, s_\ell - 1\}$ be sets, such that $\mathfrak{r}_{\ell,c}^{(k,j)}$ comes after $\mathfrak{r}_{\ell,c-1}^{(k,j)}$ in the sequence.⁵ These “sub-regime” sets indicate when coefficients $\ell \neq k$ shift within a fixed regime $\mathfrak{r}_{k,j}$. With this notation, we introduce the post-selection estimator in the next subsection.

2.2 Homogeneous case: post-CBCL estimator

As hinted in the previous subsection, we need to introduce a regressor matrix that allows the $\ell \neq k$ coefficients to break within a fixed regime set $\mathfrak{r}_{k,j}$. To do so, we define $\check{\mathbf{X}}_\ell(\mathfrak{r}_{k,j}) := \text{diag}(\mathbf{X}_\ell(\mathfrak{r}_{\ell,1}^{(k,j)}), \dots, \mathbf{X}_\ell(\mathfrak{r}_{\ell,s_\ell}^{(k,j)}))$ such that $\mathbf{X}_\ell(\mathfrak{r}_{\ell,c}^{(k,j)}) \in \mathbb{R}^{\#\mathfrak{r}_{\ell,c}^{(k,j)} \times 1}$ is variable ℓ stacking all cross-sections over dates in sub regime set $\mathfrak{r}_{\ell,c}^{(k,j)}$, where $\#A$ denotes the cardinality of any set A .⁶ With this, we define the regressor matrix $\check{\mathbf{X}}(\mathfrak{r}_{k,j})$ as

$$\check{\mathbf{X}}(\mathfrak{r}_{k,j}) := [\check{\mathbf{X}}_1(\mathfrak{r}_{k,j}) \cdots \check{\mathbf{X}}_{k-1}(\mathfrak{r}_{k,j}) \check{\mathbf{X}}_{k+1}(\mathfrak{r}_{k,j}) \cdots \check{\mathbf{X}}_p(\mathfrak{r}_{k,j}) \mathbf{F}(\mathfrak{r}_{k,j})]$$

which is a concatenation of the $\check{\mathbf{X}}_\ell(\mathfrak{r}_{k,j})$ matrices with the first time point regressors $\mathbf{F}(\mathfrak{r}_{k,j})$. Specifically, we define $\mathbf{F}(\mathfrak{r}_{k,j}) := [\mathbf{F}_1(\mathfrak{r}_{k,j}), \dots, \mathbf{F}_p(\mathfrak{r}_{k,j})]$ as a matrix of the first-time point regressors stacked in regime $\mathfrak{r}_{k,j}$ such that $\mathbf{F}_k(\mathfrak{r}_{k,j}) := \mathbf{X}_{k,1} \otimes \mathbf{1}_{\#\mathfrak{r}_{k,j}}$ with “ $\otimes$ ” being the Kronecker product and $\mathbf{1}_{\mathfrak{r}_{k,j}} \in \mathbb{R}^{\#\mathfrak{r}_{k,j} \times 1}$ being a vector of ones.⁷

Define the vectors $\tilde{\mathbf{y}}_{\mathfrak{r}_{k,j}} := [\tilde{y}_{1,T_{k,j-1}}, \dots, \tilde{y}_{1,T_{k,j}-1}, \dots, \tilde{y}_{N,T_{k,j-1}}, \dots, \tilde{y}_{N,T_{k,j}-1}]'$ and $\mathbf{X}_{k,\mathfrak{r}_{k,j}} := [x_{k,1,T_{k,j-1}}, \dots, x_{k,1,T_{k,j}-1}, \dots, x_{k,N,T_{k,j-1}}, \dots, x_{k,N,T_{k,j}-1}]'$ as the $\#\mathfrak{r}_{k,j}N \times 1$ vector forms of $\tilde{y}_{it}$ and $\mathbf{X}_{k,t}$. We can then readily show that the k th post-selection coefficient $\alpha_{k,j}$ is given by:

$$\alpha_{k,j} = \Theta_{k,\mathfrak{r}_{k,j}}^{-1} \Xi_{k,\mathfrak{r}_{k,j}}$$

where we have $\Theta_{k,\mathfrak{r}_{k,j}} := N^{-1} \mathbf{X}_{k,\mathfrak{r}_{k,j}}' \mathbf{M}_{\check{\mathbf{X}}(\mathfrak{r}_{k,j})} \mathbf{X}_{k,\mathfrak{r}_{k,j}}$ and $\Xi_{k,\mathfrak{r}_{k,j}} := N^{-1} \mathbf{X}_{k,\mathfrak{r}_{k,j}}' \mathbf{M}_{\check{\mathbf{X}}(\mathfrak{r}_{k,j})} \tilde{\mathbf{y}}_{\mathfrak{r}_{k,j}}$.

⁵Specifically, we have that $s_\ell = s_\ell^{(k,j)} := \sum_{c=1}^{m_\ell+1} \mathbb{1}(\mathfrak{r}_{\ell,c} \cap \mathfrak{r}_{k,j} \neq \emptyset)$, where $\mathbb{1}(\cdot)$ is the indicator function.

⁶Notice that $\bigcup_{c=1}^{s_\ell} \mathfrak{r}_{\ell,c}^{(k,j)} = \mathfrak{r}_{k,j}$ making it clear that we are splitting $\mathfrak{r}_{k,j}$ based on regimes of coefficient ℓ .

⁷The matrix $\check{\mathbf{X}}(\mathfrak{r}_{k,j})$ is a regressor matrix that allows breaks in the $\ell \neq k$ coefficients for dates in a fixed regime $\mathfrak{r}_{k,j}$, but also accounts for the fixed effects by including the first time as in (2.4). We refer the reader to Appendix B located in the online appendix for a full exposition on the post-selection estimator.

To retrieve all coefficients $\{\alpha_{k,j}\}$, we define $\Theta(\mathbb{T}) := \text{diag}(\Theta_1(\mathfrak{T}_{m_1}), \dots, \Theta_p(\mathfrak{T}_{m_p}))$ and $\Xi(\mathbb{T}) := [\Xi'_1(\mathfrak{T}_{m_1}), \dots, \Xi'_p(\mathfrak{T}_{m_p})]'$ with $\Theta_k(\mathfrak{T}_{m_k}) := \text{diag}(\Theta_{k,\mathfrak{r}_{k,1}}, \dots, \Theta_{k,\mathfrak{r}_{k,m_k+1}})$ and $\Xi_k(\mathfrak{T}_{m_k}) := [\Xi_{k,\mathfrak{r}_{k,1}}, \dots, \Xi_{k,\mathfrak{r}_{k,m_k+1}}]'$. Thus, the stacked vector of the true post-selection coefficients $\alpha^0(\mathbb{T}^0)$ can be conveniently expressed as $\alpha^0(\mathbb{T}^0) = \Theta(\mathbb{T}^0)^{-1} \Xi(\mathbb{T}^0)$. If we define $\hat{m} := \sum_{k=1}^p (\hat{m}_k + 1)$, we can construct the $\hat{m} \times 1$ post estimator $\hat{\alpha}(\hat{\mathbb{T}}) := [\hat{\alpha}'_1(\hat{\mathfrak{T}}_{\hat{m}_1}), \dots, \hat{\alpha}'_p(\hat{\mathfrak{T}}_{\hat{m}_p})]'$ of $\alpha^0(\mathbb{T}^0)$ as:

$$\hat{\alpha}(\hat{\mathbb{T}}) = \Theta(\hat{\mathbb{T}})^{-1} \Xi(\hat{\mathbb{T}}) \quad (2.5)$$

expressed as a function of $\hat{\mathbb{T}}$ with $\hat{\alpha}_{k,j} = \Theta_{k,\hat{\mathfrak{r}}_{k,j}}^{-1} \Xi_{k,\hat{\mathfrak{r}}_{k,j}}$ as its elements.

2.3 Heterogeneous case: penalized estimator

In this section, we discuss the case of heterogeneous breaks. Many of the notions here are similar to those in Section 2.1 so we keep the discussion brief.

We begin by assuming that cross-sections can be assigned to multiple groups, G , with $\mathcal{G} := \{1, \dots, G\}$ representing the set of these groups. For each cross-section i , let $g_i \in \mathcal{G}$ indicate the specific group to which it belongs, and let $\gamma := \{g_1, \dots, g_N\} \in \mathcal{G}^N$ be the set of such group memberships. With this setup, we consider the following DGP:

$$y_{it} = \zeta_i + \mathbf{x}'_{it} \beta_{g_i,t} + \varepsilon_{it} \quad (2.6)$$

for cross sectional units $i = 1, \dots, N$, time $t = 1, \dots, T$ and $\{g_i\}_{i=1}^N$ unknown group memberships. In the spirit of Section 2.1, we have that the k th element of $\beta_{g_i,t} := [\beta_{1,g_i,t}, \dots, \beta_{p,g_i,t}]'$, denoted as $\beta_{k,g,t}$, takes on $m_{k,g} + 1$ distinct numbers $\alpha_{k,g,j}$ such that

$$\beta_{k,g,t} = \alpha_{k,g,j} \quad (2.7)$$

for $t \in \mathfrak{r}_{k,g,j} := \{T_{k,g,j-1}, \dots, T_{k,g,j} - 1\}$, $j = 1, \dots, m_{k,g} + 1$, $m_{k,g} \in \{0, \dots, T - 1\}$, $T_{k,g,0} = 1$, and $T_{k,g,m_{k,g}+1} = T + 1$. Therefore, $\beta_{k,g,t}$ has $m_{k,g} + 1$ distinct regimes, or $m_{k,g}$ breaks, that occur at time $t \in \mathfrak{T}_{g,m_{k,g}} := \{T_{k,g,1}, \dots, T_{k,g,m_{k,g}}\}$. Now, since the break dates might differ across groups, we introduce $\mathbb{T}_k := \{\mathfrak{T}_{g,m_{k,g}}\}_{g=1}^G$, where $\mathfrak{T}_{g,m_{k,g}}$ represents the set of break dates for coefficient k of group g . With these sets, we can define $\mathbb{T} := \{\mathbb{T}_k\}_{k=1}^p$ as the full collection set of such break date collections.

As in Section 2.1, having $\mathbb{T}$ yields us the collection $\mathcal{R}_k := \{\mathfrak{r}_{k,g}\}_{g=1}^G$ with $\mathfrak{r}_{k,g} := \{\mathfrak{r}_{k,g,j}\}_{j=1}^{m_{k,g}+1}$, where $\mathfrak{r}_{k,g,j}$ is a regime set of dates within regime j in group g for coefficient k .

To estimate the true coefficients $\beta_{k,g,t}^0$, break numbers $m_{k,g}^0$, break dates $\mathbb{T}^0$, and group memberships γ^0 , we propose minimizing the following objective function:

$$\mathcal{H}_{\lambda_2}(\{\beta_{k,g,t}\}, \gamma) := \frac{1}{NT} \sum_{i=1}^N \sum_{t=2}^T \left(\tilde{y}_{it} - \mathbf{x}'_{it} \beta_{g,t} + \mathbf{x}'_{i1} \beta_{g,1} \right)^2 + \lambda_2 \sum_{g \in \mathcal{G}} \sum_{t=2}^T \ddot{\omega}'_{g,t} \Delta_{g,t}, \quad (2.8)$$

where $\tilde{y}_{it}$ is defined as in the previous section, $\lambda_2 > 0$ is a tuning parameter, $\ddot{\omega}_{g,t} := [\ddot{w}_{1,g,t}, \dots, \ddot{w}_{p,g,t}]'$ and $\Delta_{g,t} := [|\beta_{1,g,t} - \beta_{1,g,t-1}|, \dots, |\beta_{p,g,t} - \beta_{p,g,t-1}|]'$ are $p \times 1$ vectors of weights and ℓ_1 fused terms, respectively. Here, $\ddot{w}_{k,g,t} := |\ddot{\beta}_{k,g,t} - \ddot{\beta}_{k,g,t-1}|^{-\kappa_2}$ are data-driven weights with $\kappa_2 > 0$ being a user-specified parameter. Because $\ddot{\omega}_{g,t}$ and $\Delta_{g,t}$ are group-specific, we need initial *consistent* estimates of the groups, denoted as $\ddot{\gamma} := \{\ddot{g}_1, \dots, \ddot{g}_N\}$. To obtain an initial estimate of the groups, we closely follow the algorithm of Bonhomme and Manresa (2015). This is achieved by solving the following optimization problem:⁸

$$(\{\ddot{\beta}_{k,g,t}\}, \ddot{\gamma}) := \arg \min_{(\{\beta_{k,g,t}\}, \gamma)} \left\{ \sum_{i=1}^N \sum_{t=2}^T \left(\tilde{y}_{it} - \mathbf{x}'_{it} \beta_{g,t} + \mathbf{x}'_{i1} \beta_{g,1} \right)^2 \right\} \quad (2.9)$$

while keeping in mind that for a given $g \in \mathcal{G}$, we have, for all $t = 2, \dots, T$ and $k \in \mathcal{P}$, that

$$\ddot{\beta}_{k,g,t} := \left(\mathbf{X}'_{k,g,t} \mathbf{M}_{\ddot{\mathbf{X}}_{-k,g,t}} \mathbf{X}_{k,g,t} \right)^{-1} \mathbf{X}'_{k,g,t} \mathbf{M}_{\ddot{\mathbf{X}}_{-k,g,t}} \tilde{\mathbf{y}}_{g,t}, \quad (2.10)$$

where $\mathbf{X}_{k,g,t} := [x_{k,1_{g,t}}, \dots, x_{k,N_{g,t}}]'$ and $\tilde{\mathbf{y}}_{g,t} := [\tilde{y}_{1_{g,t}}, \dots, \tilde{y}_{N_{g,t}}]'$ are the $N_g \times 1$ grouped variables stacked for each time point with 1_g denoting the first cross-section in group g and $N_g := \sum_{i=1}^N \mathbb{1}\{g_i = g\}$ as the last one. Here, $\mathbb{1}(\cdot)$ is the indicator function. Further, $\mathbf{M}_{\ddot{\mathbf{X}}_{-k,g,t}} := \mathbf{I}_{N_g} - \ddot{\mathbf{X}}_{-k,g,t} (\ddot{\mathbf{X}}'_{-k,g,t} \ddot{\mathbf{X}}_{-k,g,t})^{-1} \ddot{\mathbf{X}}'_{-k,g,t}$ is defined as a projection matrix that projects out $\ddot{\mathbf{X}}_{-k,g,t} := [\mathbf{X}_{1,g,t} \cdots \mathbf{X}_{k-1,g,t} \quad \mathbf{X}_{k+1,g,t} \cdots \mathbf{X}_{p,g,t} \quad \mathbf{X}_{1,g,1} \cdots \mathbf{X}_{p,g,1}]$, an $N_g \times (2p-1)$ matrix of the $\ell \neq k$ regressors at time t in group g , augmented with the first time points.

Remark 1. When optimizing (2.8), the groups, breaks, and coefficients are estimated simultaneously. As mentioned earlier, an initial consistent estimate of the groups is required to calculate the weights, which are then used in equation (2.8) to derive the breaks. Note that these initial groups also serve as the initial inputs for the summation in the second term of (2.8). The initial estimate of the groups is obtained by minimizing (2.9), denoted as $\gamma = \ddot{\gamma}$, using the K -means algorithm described in Algorithm 2 in Appendix E. After obtaining this

⁸We note that (2.9) has not been considered in the literature yet. This represents another contribution, where we demonstrate that removing the initial period before running K -means results in the consistent estimation of groups and memberships when fixed effects are included in the model. This point does not concern breaks but is a general consideration when estimating groups.

initial estimate of $\gamma = \ddot{\gamma}$, the K -means algorithm can be applied again to update the groupings in (2.8), as outlined in Algorithm 3 in Appendix E. This process of iterating between estimating the groups and breaks can continue until convergence. It is important to emphasize that using $\gamma = \ddot{\gamma}$ as the initial estimate of the groups is crucial, as it ensures that the initial estimates in (2.10) are consistent, which are then used to estimate the weights.

Minimizing (2.8) yields the heterogeneous CBCL estimates $\{\hat{\beta}_{k,g,t}\}$, group memberships $\{\hat{g}_i\}$, and break dates $\hat{\mathbb{T}} := \{\hat{\mathbb{T}}_k\}_{k=1}^p$ with $\hat{\mathbb{T}}_k := \{\{\hat{T}_{k,1,1}, \dots, \hat{T}_{k,1,\hat{m}_{k,1}}\}, \dots, \{\hat{T}_{k,G,1}, \dots, \hat{T}_{k,G,\hat{m}_{k,G}}\}\}$, where $\hat{T}_{k,g,1} < \dots < \hat{T}_{k,g,\hat{m}_{k,g}}$ when $\hat{m}_{k,g} > 0$. We define the break dates for a given k and a group g when $|\hat{\beta}_{k,g,t} - \hat{\beta}_{k,g,t-1}| > 0$ for $t = \hat{T}_{k,g,1}, \dots, \hat{T}_{k,g,\hat{m}_{k,g}}$. If $|\hat{\beta}_{k,g,t} - \hat{\beta}_{k,g,t-1}| = 0$ for some k in group g for all $t = 1, \dots, T$, we have $\hat{m}_{k,g} = 0$ and $\hat{\mathfrak{T}}_{g,\hat{m}_{k,g}} = \mathfrak{T}_0 = \emptyset$. The estimated dates $\{\hat{\mathfrak{T}}_{g,\hat{m}_{k,g}}\}$, by definition, provide us the regime estimates $\hat{\mathfrak{r}}_{k,g,j} := \{\hat{T}_{k,g,j-1}, \dots, \hat{T}_{k,g,j} - 1\}$ of the true regime sets $\mathfrak{r}_{k,g,j}^0 := \{T_{k,g,j-1}^0, \dots, T_{k,g,j}^0 - 1\}$. Note that these sets have cardinalities $\hat{I}_{k,g,j} := |\hat{T}_{k,g,j} - \hat{T}_{k,g,j-1}|$ and $I_{k,g,j}^0 := |T_{k,g,j}^0 - T_{k,g,j-1}^0|$, respectively.

Remark 2. While the estimator for the heterogeneous case has its advantages, relying solely on it offers no free lunch, as it can be both computationally expensive and inefficient. Moreover, as a useful tip (but not a strict rule of thumb), researchers with micro-data (T fixed) may find it more efficient to use the estimator from the homogeneous case, while those with macro data ($T \rightarrow \infty$) should consider using the heterogeneous one. Note that both estimators require $N \rightarrow \infty$, as the weights rely on the cross-sectional dimension being large.

Let $\alpha^0(\mathbb{T}^0) := [\alpha_1^{0'}(\mathbb{T}_1^0), \dots, \alpha_p^{0'}(\mathbb{T}_p^0)]'$ be the true value of the post-selection vector $\alpha(\mathbb{T}) := [\alpha_1'(\mathbb{T}_1), \dots, \alpha_p'(\mathbb{T}_p)]'$, where $\alpha_k(\mathbb{T}_k) := [\alpha_{k,1}'(\mathfrak{T}_{1,m_{k,1}}), \dots, \alpha_{k,G}'(\mathfrak{T}_{G,m_{k,G}})]'$ such that we have $\alpha_{k,g} := [\alpha_{k,g,1}, \dots, \alpha_{k,g,m_{k,g}+1}]'$ being the k th post-selection vector in group $g \in \mathcal{G}$. To estimate the coefficients $\{\alpha_{k,g,j}^0\}$, we introduce sub-regime sets as in Section 2.1. For any $\ell \neq k$, consider the sequence of sets $\mathfrak{r}_{\ell,g,1} \cap \mathfrak{r}_{k,g,j}, \dots, \mathfrak{r}_{\ell,g,m_{\ell,g}+1} \cap \mathfrak{r}_{k,g,j}$. Define $\mathfrak{r}_{\ell,g,1}^{(k,j)}$ as the first instance of a non-empty set in the sequence, $\mathfrak{r}_{\ell,g,s_{\ell,g}}^{(k,j)}$ as the last instance of a non-empty set, and $\mathfrak{r}_{\ell,g,c}^{(k,j)}$ as sets for $c \in \{2, \dots, s_{\ell,g} - 1\}$, such that $\mathfrak{r}_{\ell,g,c}^{(k,j)}$ comes after $\mathfrak{r}_{\ell,g,c-1}^{(k,j)}$ in the sequence. With this notation, we introduce the post-selection estimator in the next subsection.

2.4 Heterogeneous case: post-CBCL estimator

Define $\check{\mathbf{X}}_{\ell,g}(\mathfrak{r}_{k,g,j}) := \text{diag}(\mathbf{X}_{\ell,g}(\mathfrak{r}_{\ell,g,1}^{(k,j)}), \dots, \mathbf{X}_{\ell,g}(\mathfrak{r}_{\ell,g,s_{\ell,g}}^{(k,j)}))$ such that $\mathbf{X}_{\ell,g}(\mathfrak{r}_{\ell,g,c}^{(k,j)}) \in \mathbb{R}^{\#\mathfrak{r}_{\ell,g,c}^{(k,j)} N_g \times 1}$ is variable ℓ stacking cross-sections in group $g \in \mathcal{G}$ over dates in regime set $\mathfrak{r}_{\ell,g,c}^{(k,j)}$. Further,

define $\mathbf{F}_g(\mathbf{r}_{k,g,j}) := [\mathbf{F}_{1,g}(\mathbf{r}_{k,g,j}), \dots, \mathbf{F}_{p,g}(\mathbf{r}_{k,g,j})]$ as the first time point regressors in group g such that $\mathbf{F}_{k,g}(\mathbf{r}_{k,g,j}) := \mathbf{X}_{k,g,1} \otimes \mathbf{1}_{\#\mathbf{r}_{k,g,j}}$. With this, we introduce:

$$\check{\mathbf{X}}_g(\mathbf{r}_{k,g,j}) := [\check{\mathbf{X}}_{1,g}(\mathbf{r}_{k,g,j}) \cdots \check{\mathbf{X}}_{k-1,g}(\mathbf{r}_{k,g,j}) \check{\mathbf{X}}_{k+1,g}(\mathbf{r}_{k,g,j}) \cdots \check{\mathbf{X}}_{p,g}(\mathbf{r}_{k,g,j}) \mathbf{F}_g(\mathbf{r}_{k,g,j})],$$

a regressor matrix that allows breaks in the $\ell \neq k$ coefficients for dates in a fixed regime set $\mathbf{r}_{k,g,j}$. To derive the post-selection coefficient, we define $\tilde{\mathbf{y}}_{g,\mathbf{r}_{k,g,j}}$ and $\mathbf{X}_{k,g,\mathbf{r}_{k,g,j}}$ as the $\#\mathbf{r}_{k,g,j}N_g \times 1$ stacked vectors of $\tilde{y}_{it}$ and $\mathbf{X}_{k,g,t}$, respectively, for dates in set $\mathbf{r}_{k,g,j}$ and cross-sections in $g \in \mathcal{G}$. If we further define $\Psi_{k,g,\mathbf{r}_{k,g,j}} := (N_g I_{k,g,j})^{-1} \mathbf{X}'_{k,g,\mathbf{r}_{k,g,j}} \mathbf{M}_{\check{\mathbf{X}}_g(\mathbf{r}_{k,g,j})} \mathbf{X}_{k,g,\mathbf{r}_{k,g,j}}$ and $\mathbf{Y}_{k,g,\mathbf{r}_{k,g,j}} := (N_g I_{k,g,j})^{-1} \mathbf{X}'_{k,g,\mathbf{r}_{k,g,j}} \mathbf{M}_{\check{\mathbf{X}}_g(\mathbf{r}_{k,g,j})} \tilde{\mathbf{y}}_{g,\mathbf{r}_{k,g,j}}$, we can show that the k th post-selection coefficient in regime j of group g is given by $\alpha_{k,g,j} = \Psi_{k,g,\mathbf{r}_{k,g,j}}^{-1} \mathbf{Y}_{k,g,\mathbf{r}_{k,g,j}}$.

Remark 3. The post-selection estimator here is similar to that in Section 2.2, but it is computed separately for each group $g \in \mathcal{G}$. Another difference is that, since $T \rightarrow \infty$ and each coefficient is derived from distinct regime blocks, the asymptotics depend on the size of each coefficient-specific regime, $I_{k,g,j} = \#\mathbf{r}_{k,g,j}$, which may differ across coefficients.

To retrieve all of the coefficients $\{\alpha_{k,g,j}\}$, we define the full block diagonal matrix $\Psi(\mathbb{T}) := \text{diag}(\Psi_1(\mathbb{T}_1), \dots, \Psi_p(\mathbb{T}_p))$ and the full vector $\mathbf{Y}(\mathbb{T}) := [\mathbf{Y}'_1(\mathbb{T}_1), \dots, \mathbf{Y}'_p(\mathbb{T}_p)]'$, where we have that $\Psi_k(\mathbb{T}_k) := \text{diag}(\Psi_{k,1}(\mathfrak{T}_{1,m_{k,1}}), \dots, \Psi_{k,G}(\mathfrak{T}_{G,m_{k,G}}))$ being the k th block diagonal of $\Psi(\mathbb{T})$ and $\mathbf{Y}_k(\mathbb{T}_k) := [\mathbf{Y}'_{k,1}(\mathfrak{T}_{1,m_{k,1}}), \dots, \mathbf{Y}'_{k,G}(\mathfrak{T}_{G,m_{k,G}})]'$ as the k th vector of $\mathbf{Y}(\mathbb{T})$. Further, the g th component in the k th blocks $\Psi_k(\mathbb{T}_k)$ and $\mathbf{Y}_k(\mathbb{T}_k)$ are given by $\Psi_{k,g} := \text{diag}(\Psi_{k,g,\mathbf{r}_{k,g,1}}, \dots, \Psi_{k,g,\mathbf{r}_{k,g,m_{k,g}+1}})$ and $\mathbf{Y}_{k,g} := [\mathbf{Y}_{k,g,\mathbf{r}_{k,g,1}}, \dots, \mathbf{Y}_{k,g,\mathbf{r}_{k,g,m_{k,g}+1}}]'$, respectively. These elements are analogous to those in Section 2.2, but with the caveat of accounting for the group structures. If we define $\hat{m} := \sum_{k=1}^p \sum_{g=1}^G (\hat{m}_{k,g} + 1)$, we can construct the $\hat{m} \times 1$ estimator $\hat{\mathbf{a}}(\hat{\mathbb{T}}) := [\hat{\mathbf{a}}'_1(\hat{\mathbb{T}}_1), \dots, \hat{\mathbf{a}}'_p(\hat{\mathbb{T}}_p)]'$ of $\alpha^0(\mathbb{T}^0)$ given by $\hat{\mathbf{a}}(\hat{\mathbb{T}}) = \Psi(\hat{\mathbb{T}})^{-1} \mathbf{Y}(\hat{\mathbb{T}})$ expressed as a function of $\hat{\mathbb{T}}$.

3 Asymptotic results

We work with Assumptions 1.1 to 1.4 for the homogeneous estimator and 2.1 to 2.4 for the heterogeneous estimator. We first introduce some notations. If $\mathbf{A}$ is a matrix, $\mu_{\min}(\mathbf{A})$ and $\mu_{\max}(\mathbf{A})$ denote its smallest and largest singular values, respectively, $\text{tr } \mathbf{A}$ denotes its trace, and $\|\mathbf{A}\| := \sqrt{\text{tr } \mathbf{A}' \mathbf{A}}$ represents its Frobenius norm. The terms w.p.1, plim, $\rightarrow_d$, $\rightarrow_p$, $\mathcal{N}(\cdot, \cdot)$, and $\mathbf{0}_{r \times 1}$ signify probability one, probability limit, convergence in distribution, convergence in probability, a normal distribution, and an $r \times 1$ vector of zeros, respectively.

3.1 Homogeneous case: assumptions

Before outlining the conditions, we define the vector $\mathbf{U}(\mathbb{T}^0) := [\mathbf{U}'_1(\mathfrak{T}_{m_1^0}), \dots, \mathbf{U}'_p(\mathfrak{T}_{m_p^0})]'$ such that the j th element of $\mathbf{U}_k(\mathfrak{T}_{m_k^0}) \in \mathbb{R}^{(m_k^0+1) \times 1}$ is given by $U_{k,r_{k,j}^0} := N^{-1} \mathbf{X}'_{k,r_{k,j}^0} \mathbf{M}_{\check{\mathbf{x}}(r_{k,j}^0)} \tilde{\mathbf{e}}_{r_{k,j}^0}$ with $\tilde{\mathbf{e}}_{r_{k,j}^0}$ being the $\#r_{k,j}^0 \cdot N \times 1$ stacked vector of $\tilde{\varepsilon}_{it}$, where $\tilde{\varepsilon}_{it} := \varepsilon_{it} - \varepsilon_{i1}$.

Assumption 1.1. We assume that $\mathbb{E}(\varepsilon_{it} \mathbf{x}_{is}) = 0$ for all i, t and s . Further, we assume that $\mathbb{E}(\|\mathbf{x}_{it}\|^4) < \infty$, $\mathbb{E}(\varepsilon_{it}^4) < \infty$ for all i and t , and that $\sup_t N^{-1} \sum_{i=1}^N \sum_{j=1}^N |\mathbb{E}(\tilde{\varepsilon}_{it} \mathbf{x}'_{it} \mathbf{x}_{jt} \tilde{\varepsilon}_{jt})| < \infty$ and $\sup_{t,s} N^{-1} \sum_{i=1}^N \sum_{j=1}^N |\mathbb{E}(\tilde{\varepsilon}_{it} \mathbf{x}'_{i1} \mathbf{x}_{j1} \tilde{\varepsilon}_{js})| < \infty$ for any N .

Assumption 1.2.

- (a) $\inf_t \mu_{\min} \left(N^{-1} \sum_{i=1}^N \mathbf{x}_{it} \mathbf{x}'_{it} \right) > 0$ w.p.1;
- (b) $\boldsymbol{\Theta}(\mathbb{T}) \rightarrow_p \boldsymbol{\Theta}_0(\mathbb{T}) := \lim_{N \rightarrow \infty} \mathbb{E}[\boldsymbol{\Theta}(\mathbb{T})]$ as $N \rightarrow \infty$, where $\infty > \inf_{\mathbb{T}} \mu_{\min}[\boldsymbol{\Theta}_0(\mathbb{T})] > 0$;
- (c) $\sqrt{N} \mathbf{U}(\mathbb{T}^0) \rightarrow_d \mathcal{N} \left(\mathbf{0}_{\sum_{k=1}^p (m_k^0+1) \times 1}, \boldsymbol{\Sigma}_0 \right)$.

Assumption 1.3.

- (a) $J_{1,\max} := \max_{\{1 \leq j \leq m_k^0+1, 1 \leq k \leq p\}} |\alpha_{k,j+1}^0 - \alpha_{k,j}^0| = O(1)$;
- (b) $\sqrt{N} \lambda_1 J_{1,\min}^{-\kappa_1} \rightarrow_p c_1 \in [0, \infty)$ and $\sqrt{N} J_{1,\min} \rightarrow \infty$ as $N \rightarrow \infty$, where we define $J_{1,\min} := \min_{\{1 \leq j \leq m_k^0+1, 1 \leq k \leq p\}} |\alpha_{k,j+1}^0 - \alpha_{k,j}^0|$;
- (c) $N^{(\kappa_1+1)/2} \lambda_1 \rightarrow_p \infty$ as $N \rightarrow \infty$.

Assumption 1.4. $T \geq 2$ is fixed as $N \rightarrow \infty$.

Before discussing the asymptotic properties of the homogeneous estimator, we address some key points. We keep the discussion concise, as many of these Assumptions are standard in the literature (see also Qian and Su 2016; and Kaddoura and Westerlund 2023). In Assumption 1.1, we assume that the errors and regressors are independent, that the fourth-order moments are finite, and that there is at most weak cross-sectional correlation. Lagged dependent variables can be permitted, but in that case, ε_{it} cannot be serially correlated, nor can the fixed effects term ζ_i be included. Assumption 1.1 can be relaxed if proper instrumental variables are available, by using a GMM-type penalized objective, as demonstrated in studies such as Qian and Su (2016). Furthermore, the heteroscedasticity properties of ε_{it} are not restricted, and similarly, the time-series dependence is not constrained, as T is finite.

Assumption 1.2 implies that the projected regressors are noncollinear. Assumption 1.2(c) is commonly adopted in the literature, so we will follow this convention as well (see Qian and Su 2016). This Assumption is not stringent and can be satisfied by various mixing processes (see Bai 2003). In Assumption 1.4, we emphasize that T is fixed as $N \rightarrow \infty$.

In Assumption 1.3, we establish conditions on the tuning parameter λ_1 and the magnitude of differences between regimes. This Assumption parallels the one made by Qian and Su (2016), to which we refer for further discussion. One difference worth highlighting is that since $J_{1,\min}$ represents the minimum overall coefficient magnitude differences, it is possible to have fixed jumps in some parameters and shrinking ones in others. This is evident by the fact that $\sqrt{N}J_{1,\min} \rightarrow \infty$ is satisfied if $J_{1,\min}$ is constant and does not depend on N , but also if $J_{1,\min}$ shrinks to zero, provided that it does not at a rate that is slower than $N^{-1/2}$. Furthermore, we note that $\kappa_1 > 0$ is a positive fixed constant incorporated into the weights, $w_{k,t} = |\dot{\beta}_{k,t} - \dot{\beta}_{k,t-1}|^{-\kappa_1}$ and is typically set to $\kappa_1 = 2$ in practice (see, e.g., Qian and Su 2016; Okui and Wang 2021; and Kaddoura and Westerlund 2024).

3.2 Homogeneous case: asymptotics

Our first main result characterizes the limit of $\hat{\beta}_{k,t}$.

Theorem 1.1. *Suppose that Assumptions 1.1-1.4 hold. Then, uniformly in $k = 1, \dots, p$ and $t \in \mathfrak{T}_{T-1}$, we have that $\hat{\beta}_{k,t} - \beta_{k,t}^0 = O_p(N^{-1/2})$.*

In Theorem 1.1, we establish that the estimator is $\sqrt{N}$ consistent. This implies we can achieve “selection consistency” about the number of breaks and the dates of the breaks, which leads us to the next theorem.

Theorem 1.2. *Suppose that Assumptions 1.1-1.4 hold. Then, as $N \rightarrow \infty$,*

$$\mathbb{P} \left(\left\{ \left| \hat{\beta}_{k,t} - \hat{\beta}_{k,t-1} \right| = 0 \forall t \in \mathfrak{T}_{m_k^0}^{0c} \text{ and } k \in \mathcal{P} \right\} \right) \rightarrow 1,$$

where $\mathfrak{T}_{m_k^0}^{0c} := \{2, \dots, T\} \setminus \mathfrak{T}_{m_k^0}^0$.

Theorem 1.2 demonstrates that $\hat{\beta}_{k,t} - \hat{\beta}_{k,t-1}$ is consistent for $\beta_{k,t}^0 - \beta_{k,t-1}^0$ when $t \in \mathfrak{T}_{m_k^0}^{0c}$ for all $k = 1, \dots, p$. However, Theorem 1.1 states that $\hat{\beta}_{k,t} - \hat{\beta}_{k,t-1}$ is consistent for all k and t including $t \in \mathfrak{T}_{m_k^0}^0$. This implies that the procedure can correctly identify the true model,

along with the correct number of breaks and their dates. In the following corollary, we formalize this.

Corollary 1.1 *Suppose that Assumptions 1.1-1.4 hold. Then, as $N \rightarrow \infty$,*

1. $\mathbb{P}(\{\hat{m}_k = m_k^0 \forall k\}) \rightarrow 1$;
2. $\mathbb{P}\left(\left\{\widehat{\mathfrak{T}}_{\hat{m}_k} = \mathfrak{T}_{m_k^0}^0 \forall k \mid \hat{m}_k = m_k^0 \forall k\right\}\right) \rightarrow 1$.

Corollary 1.1 implies that our approach can identify the true model with the correct number of breaks and break dates, which consequently implies, by definition, that $\widehat{\mathfrak{t}}_{k,j} = \mathfrak{t}_{k,j}^0$ for all k and j with probability approaching one. In Theorem 1.3, we report the asymptotic distribution of the CBCL post-selection estimator.

Theorem 1.3. *Suppose that Assumptions 1.1-1.4. Then, conditional on $\hat{m}_k = m_k^0$ for all $k = 1, \dots, p$, as $N \rightarrow \infty$,*

$$\sqrt{N}(\widehat{\alpha} - \alpha^0) \rightarrow_d \mathcal{N}\left(\mathbf{0}_{\sum_{k=1}^p (m_k^0 + 1) \times 1}, \mathbf{\Theta}_0^{-1} \Sigma_0 \mathbf{\Theta}_0^{-1}\right).$$

The fact that the above theorem conditions on the high probability event $\{\hat{m}_k = m_k^0 \forall k\}$ shows that the CBCL estimates are oracle efficient, i.e., asymptotically equivalent to the infeasible LS estimator that takes all breaks as known. Furthermore, since the estimator is asymptotically normal, it supports standard normal and chi-squared inference.

3.2.1 Choosing the tuning parameter λ_1

The estimator discussed in previous sections depends on the tuning parameter λ_1 . For a comprehensive explanation on selecting λ_1 in practice, we refer to Appendix E. We select λ_1 in a data-driven manner, similar to the approaches taken in prior studies such as Qian and Su (2016), Okui and Wang (2021), and Kaddoura and Westerlund (2023), by minimizing the following information criterion:

$$\text{IC}_1(\lambda_1) := \sigma_{\lambda_1}^2 \left(\widehat{\mathbb{T}}(\lambda_1) \right) + \varphi_1 \cdot \left[\sum_{k=1}^p (\widehat{m}_k(\lambda_1) + 1) \right], \quad (3.1)$$

where $\widehat{\mathbb{T}}(\lambda_1)$ and $\widehat{m}_k(\lambda_1)$ are $\widehat{\mathbb{T}}$ and $\widehat{m}_k$, respectively, viewed as functions of λ_1 with $\varphi_1 = \varphi_1(N) > 0$ being a penalty. By fixing any $k \in \mathcal{P}$, we define $\sigma_{\lambda_1}^2(\widehat{\mathbb{T}}(\lambda_1))$ as:

$$\frac{1}{N} \sum_{i=1}^N \sum_{j=1}^{\widehat{m}_k(\lambda_1)+1} \sum_{t=\widehat{T}_{k,j-1}(\lambda_1)}^{\widehat{T}_{k,j}(\lambda_1)-1} \left(\widetilde{y}_{it} - x_{k,it} \widehat{\alpha}_{k,j} + x_{k,i1} \widehat{\alpha}_{k,1} - \sum_{\ell \neq k} [x_{\ell,it} \widehat{\phi}_{\ell,t} - x_{\ell,i1} \widehat{\phi}_{\ell,1}] \right)^2.$$

Here, $\hat{\phi}_{\ell,t} := \sum_{j'=1}^{\hat{m}_\ell+1} \mathbb{1}(t \in \hat{\tau}_{\ell,j'}) \hat{\alpha}_{\ell,j'}$ represents a coefficient that assigns the appropriate post-selection estimator for each $\ell \neq k$ variable, where recall that $\hat{\tau}_{\ell,j'} = \{\hat{T}_{\ell,j'-1}, \dots, \hat{T}_{\ell,j'} - 1\}$. The penalty φ_1 is not unique and must be set by the researcher. To obtain the true model by minimizing equation (3.1), we require that $\varphi_1 \rightarrow 0$ and $N\varphi_1 \rightarrow \infty$. Therefore, we recommend using $\varphi_1(N) = c[\log(N)/\sqrt{N}]$, with $c = 0.05$, as in Qian and Su (2016). Similar to Qian and Su (2016), our approach is not sensitive to reasonable variations in the constant c . The choice of φ_1 makes the information criterion similar to the conventional Schwarz-Bayesian information criterion (BIC). Theorem 1.4 in Appendix C shows that minimizing IC_1 leads to selecting an appropriate λ_1 giving us a consistent estimate of the breaks for all coefficients.

3.3 Heterogeneous case: assumptions

First, we introduce more notation. Let $\hat{m}^0 := \sum_{k=1}^p \sum_{g=1}^G (m_{k,g}^0 + 1)$, and define the $\ell \times \hat{m}^0$ selection matrix $\mathbf{S}$ such that $\|\mathbf{S}\|$ is finite, where $\ell \in [1, \hat{m}^0]$ is a fixed integer. Let $N_g/N \rightarrow \pi_g$ and define $\mathbf{\Pi} := \text{diag}(\mathbf{\Pi}_1, \dots, \mathbf{\Pi}_p)$, where $\mathbf{\Pi}_k := \text{diag}(\mathbf{\Pi}_{k,1}, \dots, \mathbf{\Pi}_{k,G})$ with $\mathbf{\Pi}_{k,g} \in \mathbb{R}^{(m_{k,g}^0+1) \times (m_{k,g}^0+1)}$ being a diagonal matrix of π_g as the diagonal values. Next, we introduce the diagonal matrix $\mathbf{O}_{k,g} = \mathbf{O}_{k,g}(\mathfrak{T}_{g,m_{k,g}}) := \text{diag}(I_{k,g,1}, \dots, I_{k,g,m_{k,g}+1})$, where $I_{k,g,j} := |T_{k,g,j} - T_{k,g,j-1}|$ represents an interval length of a certain regime. Now, denote $\mathbf{O}(\mathbb{T}) := \text{diag}(\mathbf{O}_1(\mathbb{T}_1), \dots, \mathbf{O}_p(\mathbb{T}_p))$ with $\mathbf{O}_k(\mathbb{T}_k) := \text{diag}(\mathbf{O}_{k,1}, \dots, \mathbf{O}_{k,G})$. With this setup, we define $\mathbf{O}_0 := \mathbf{O}(\mathbb{T}^0) \in \mathbb{R}^{\hat{m}^0 \times \hat{m}^0}$ and proceed to consider Assumptions 2.1–2.4 based on this notation.

Assumption 2.1.

- (a) $\mathbb{B}$ is a compact subset of $\mathbb{R}^{p \times 1}$, where $\mathbb{B}$ is the parameter set of each $\beta_{g,t}$;
- (b) We assume that $\mathbb{E}(\varepsilon_{it}\mathbf{x}_{is}) = 0$ for all i, t and s . Further, we assume $\mathbb{E}(\|\mathbf{x}_{it}\|^4) < \infty$ and $\mathbb{E}(\varepsilon_{it}^4) < \infty$ for all i and t . We also assume $N^{-1} \sum_{i=1}^N \sum_{j=1}^N |T^{-1} \sum_{t=2}^T \mathbb{E}(\tilde{\varepsilon}_{it}\mathbf{x}'_{it}\mathbf{x}_{jt}\tilde{\varepsilon}_{jt})| < \infty$ and $N^{-1} \sum_{i=1}^N \sum_{j=1}^N |T^{-1} \sum_{t=2}^T \mathbb{E}(\tilde{\varepsilon}_{it}\mathbf{x}'_{i1}\mathbf{x}_{j1}\tilde{\varepsilon}_{jt})| < \infty$ for any N and T ;
- (d) For any N and T , we have $\left| N^{-2} \sum_{i=1}^N \sum_{j=1}^N T^{-1} \sum_{t=2}^T \sum_{s=2}^T \text{Cov}(\tilde{\varepsilon}_{it}\mathbf{x}'_{it}\mathbf{x}_{jt}\tilde{\varepsilon}_{jt}, \tilde{\varepsilon}_{is}\mathbf{x}'_{is}\mathbf{x}_{js}\tilde{\varepsilon}_{js}) \right| < \infty$ and $\left| N^{-2} \sum_{i=1}^N \sum_{j=1}^N T^{-1} \sum_{t=2}^T \sum_{s=2}^T \text{Cov}(\tilde{\varepsilon}_{it}\mathbf{x}'_{i1}\mathbf{x}_{j1}\tilde{\varepsilon}_{jt}, \tilde{\varepsilon}_{is}\mathbf{x}'_{i1}\mathbf{x}_{j1}\tilde{\varepsilon}_{js}) \right| < \infty$.

Assumption 2.2.

- (a) Define $\hat{\mu}(\gamma, g, \tilde{g}) := \inf_t \mu_{\min} \left(N^{-1} \sum_{i=1}^N \mathbb{1}\{g_i^0 = g\} \mathbb{1}\{g_i = \tilde{g}\} \mathbf{x}_{it}\mathbf{x}'_{it} \right)$. There exists $\hat{\mu}$ and $\mu > 0$ such that $\hat{\mu} \rightarrow_p \mu$ and $\forall g \in \mathcal{G}, \min_{\gamma \in \mathcal{G}^N} \max_{\tilde{g} \in \mathcal{G}} \hat{\mu}(\gamma, g, \tilde{g}) > \hat{\mu}$;
- (b) $\Psi(\mathbb{T}) \rightarrow_p \Psi_0(\mathbb{T}) := \lim_{N,T \rightarrow \infty} \mathbb{E}[\Psi(\mathbb{T})]$ as $N, T \rightarrow \infty$, where $\infty > \inf_{\mathbb{T}} \mu_{\min}[\Psi_0(\mathbb{T})] > 0$;

- (c) $\sqrt{N}\mathbf{S}\mathbf{\Psi}_0^{-1}\mathbf{O}_0^{1/2}\mathbf{W}(\mathbb{T}^0) \rightarrow_d \mathcal{N}(\mathbf{0}, \mathbf{S}\mathbf{\Psi}_0^{-1}\mathbf{\Pi}^{-1/2}\mathbf{\Lambda}_0\mathbf{\Pi}^{-1/2}\mathbf{\Psi}_0^{-1}\mathbf{S}')$, where $\mathbf{W}(\mathbb{T}^0)$ is defined similarly to $\mathbf{Y}(\mathbb{T}^0)$, but it consists of $(N_g I_{k,g,j}^0)^{-1} \mathbf{X}'_{k,g,\tau_{k,g,j}^0} \mathbf{M}_{\tilde{\mathbf{x}}_g(\tau_{k,g,j}^0)} \tilde{\mathbf{\epsilon}}_{g,\tau_{k,g,j}^0}$ as elements;
- (d) Let $D_{g\tilde{g}i} := T^{-1} \sum_{t=2}^T \left(\mathbf{x}'_{it}(\boldsymbol{\beta}_{g,i,t}^0 - \boldsymbol{\beta}_{\tilde{g},i,t}^0) - \mathbf{x}'_{i1}(\boldsymbol{\beta}_{g,i,1}^0 - \boldsymbol{\beta}_{\tilde{g},i,1}^0) \right)^2$. For all $g \neq \tilde{g}$, there exists $c_{g\tilde{g}} > 0$ such that $\text{plim}_{N,T \rightarrow \infty} N^{-1} \sum_{i=1}^N D_{g\tilde{g}i} > c_{g\tilde{g}}$ and for all i , $\text{plim}_{T \rightarrow \infty} D_{g\tilde{g}i} > c_{g\tilde{g}}$.

Assumption 2.3.

- (a) There exists two constants $R_{\tilde{\epsilon},\mathbf{x}}$ and $R_{\mathbf{x}}$ such that as $N, T \rightarrow \infty$ and $\delta > 0$, the following hold: $\sup_i \mathbb{P}(T^{-1} \sum_{t=2}^T \|\tilde{\epsilon}_{it} \mathbf{x}_{it}\|^2 \geq R_{\tilde{\epsilon},\mathbf{x}})$, $\sup_i \mathbb{P}(T^{-1} \sum_{t=2}^T \|\tilde{\epsilon}_{it} \mathbf{x}_{i1}\|^2 \geq R_{\tilde{\epsilon},\mathbf{x}})$, and $\sup_i \mathbb{P}(T^{-1} \sum_{t=2}^T \|\mathbf{x}_{it}\|^4 \geq R_{\mathbf{x}})$ are $o(T^{-\delta})$;
- (b) There exists constants $a > 0$ and $d_1 > 0$ and a sequence $\alpha[t] \leq e^{-at^{d_1}}$ such that for all $i = 1, \dots, N$, $(g, \tilde{g}) \in \mathcal{G}^2$ and $g \neq \tilde{g}$, $\{\mathbf{x}'_{it}(\boldsymbol{\beta}_{g,i,t}^0 - \boldsymbol{\beta}_{\tilde{g},i,t}^0) - \mathbf{x}'_{i1}(\boldsymbol{\beta}_{g,i,1}^0 - \boldsymbol{\beta}_{\tilde{g},i,1}^0)\}_t$ and $\{(\mathbf{x}'_{it}(\boldsymbol{\beta}_{g,i,t}^0 - \boldsymbol{\beta}_{\tilde{g},i,t}^0) - \mathbf{x}'_{i1}(\boldsymbol{\beta}_{g,i,1}^0 - \boldsymbol{\beta}_{\tilde{g},i,1}^0))\tilde{\epsilon}_{it}\}_t$ are strongly mixing processes with mixing coefficients $\alpha[t]$. Further, we have that $\mathbb{E}(\mathbf{x}'_{it}(\boldsymbol{\beta}_{g,i,t}^0 - \boldsymbol{\beta}_{\tilde{g},i,t}^0)\tilde{\epsilon}_{it}) = 0$ and $\mathbb{E}(\mathbf{x}'_{i1}(\boldsymbol{\beta}_{g,i,1}^0 - \boldsymbol{\beta}_{\tilde{g},i,1}^0)\tilde{\epsilon}_{it}) = 0$;
- (c) There exists constants $b_x > 0, b_e > 0, d_{2x} > 0$ and d_{2e} such that $\mathbb{P}(|\mathbf{x}'_{it}(\boldsymbol{\beta}_{g,i,t}^0 - \boldsymbol{\beta}_{\tilde{g},i,t}^0) - \mathbf{x}'_{i1}(\boldsymbol{\beta}_{g,i,1}^0 - \boldsymbol{\beta}_{\tilde{g},i,1}^0)| > m) \leq e^{1-(m/b_x)^{d_{2x}}}$ and $\mathbb{P}(|(\mathbf{x}'_{it}(\boldsymbol{\beta}_{g,i,t}^0 - \boldsymbol{\beta}_{\tilde{g},i,t}^0) - \mathbf{x}'_{i1}(\boldsymbol{\beta}_{g,i,1}^0 - \boldsymbol{\beta}_{\tilde{g},i,1}^0))\tilde{\epsilon}_{it}| > m) \leq e^{1-(m/b_e)^{d_{2e}}}$ for any i, t , and $m > 0$.

Assumption 2.4.

- (a) $J_{2,\max} := \max_{\{1 \leq g \leq G, 1 \leq j \leq m_{k,g}^0 + 1, 1 \leq k \leq p\}} |\alpha_{k,g,j+1}^0 - \alpha_{k,g,j}^0| = O(1)$;
- (b) $\sqrt{NT\hat{m}^0}\lambda_2 J_{2,\min}^{-\kappa_2} \rightarrow_p c_1 \in [0, \infty)$ and $\sqrt{N}J_{2,\min} \rightarrow \infty$ as $N, T \rightarrow \infty$, where $J_{2,\min} := \min_{\{1 \leq g \leq G, 1 \leq j \leq m_{k,g}^0 + 1, 1 \leq k \leq p\}} |\alpha_{k,g,j+1}^0 - \alpha_{k,g,j}^0|$;
- (c) $TN^{(\kappa_2+1)/2}\lambda_2 \rightarrow_p \infty$ as $N, T \rightarrow \infty$.

Aside from Assumption 2.3, most of these assumptions are similar to those in the homogeneous section, and hence we will keep this discussion brief. Assumption 2.1 is essentially the same as that given by Assumption 1.1. In Assumption 2.2, we assume noncollinearity in the regressors. Assumptions 2.2(b)-(d) are typical in the literature on groups and breaks (see, for example, Qian and Su 2016; and Okui and Wang 2021). For instance, Assumption 2.2(c) ensures asymptotic normality with a convergence rate of $\sqrt{N_g I_{k,g,j}}$, an assumption

that is also utilized in, for example, Okui and Wang (2021). This also highlights that estimating breaks (and, consequently, the regimes) for each coefficient, could yield asymptotic improvements since the convergence rates depend on the coefficient-specific regimes characterized by the sample size $I_{k,g,j}$. In Assumption 2.2(d), we restrict the difference between the predicted values of two groups for a certain cross-section i . The condition states that this difference is bounded away from zero (see Bonhomme and Manresa 2015, for a discussion). In Assumption 2.3, we set conditions on the tail behavior of the variables and the dependence structure. For example, we do not allow the tails of the distribution of the regressors to lack a finite fourth moment, nor do we allow unit roots in the regressors. In Assumption 2.3(b)-(d), we assume that the mixing coefficient $\alpha[t]$ decays exponentially; this assumption is made to apply the exponential inequalities utilized in Bonhomme and Manresa (2015). Finally, Assumption 2.4 is analogous to Assumption 1.3. However, because $T \rightarrow \infty$, the number of breaks, $\dot{m}^0$, may diverge, albeit at a slow rate.

3.4 Heterogeneous case: asymptotics

Our first asymptotic result is presented in Proposition 2.1 which establishes the $\sqrt{N}$ consistency of the initial estimators $\ddot{\beta}_{k,g,t}$. This, in turn, ensures the consistency of our main estimator, as it depends on the weights, $\ddot{w}_{k,g,t} = |\ddot{\beta}_{k,g,t} - \ddot{\beta}_{k,g,t-1}|^{-\kappa_2}$. Intuitively, this guarantees that the penalty is adaptive in the sense that, for all groups, the weights become large when there is no break and small when there is one.

Proposition 2.1. *Consider that Assumptions 2.1-2.3 hold. Also, suppose that $N_g/N \rightarrow \pi_g \in (0, 1)$ for any $g \in \mathcal{G}$ and $N/T^\delta \rightarrow 0$ for any $\delta > 0$. Then, for all k, g and t , $\ddot{\beta}_{k,g,t} - \beta_{k,g,t}^0 = o_p(N^{-1/2})$.*

Before presenting the next theorem, we introduce $\beta_{k,g,t}^\dagger$, which represents the argument that minimizes $\mathcal{H}(\{\beta_{k,g,t}\}) := \mathcal{H}(\{\beta_{k,g,t}\}, \gamma^0)$. The objective function $\mathcal{H}(\{\beta_{k,g,t}\})$ assumes that group memberships are known, leaving us with the task of estimating the number of breaks, denoted by $m_{k,g}^\dagger$, and the dates, denoted by $\mathbb{T}^\dagger$. This brings us to Theorem 2.1.

Theorem 2.1. *Suppose that Assumptions 2.1-2.3 are satisfied and that $N_g/N \rightarrow \pi_g \in (0, 1)$ for all $g \in \mathcal{G}$. As $N, T \rightarrow \infty$, for any $\delta > 0$, it holds, for all k, g and t , that, $\ddot{\beta}_{k,g,t} = \beta_{k,g,t}^\dagger + o_p(T^{-\delta})$.*

Note that δ in Theorem 2.1 can be arbitrarily large, in the sense that $\delta \gg 1$, which implies that the estimator, without the knowledge of group membership, is "super consistent" with

the rate of convergence $T^{-\delta}$. This is crucial because we show that the error coming from group estimation is negligible, as the difference between $\hat{\beta}_{k,g,t}$ and $\beta_{k,g,t}^+$ is small. This implies that we can accurately detect the number of breaks and break dates without knowing the group memberships. In the next theorem, we show that $\beta_{k,g,t}^+$ has a $\sqrt{N}$ convergence rate.

Theorem 2.2. *Suppose that Assumptions 2.1-2.4 hold and $N_g/N \rightarrow \pi_g \in (0,1)$ for any $g \in \mathcal{G}$. Then, as $N, T \rightarrow \infty$, we have that $\beta_{k,g,t}^+ - \beta_{k,g,t}^0 = O_p(N^{-1/2})$ for all k, g and t .*

This result, combined with the result of Theorem 2.1, i.e., $\hat{\beta}_{k,g,t} - \beta_{k,g,t}^+ = o_p(T^{-\delta})$, demonstrates the consistency of the shrinkage estimator. In fact, the proof of Theorem 2.2, which asserts that the procedure can correctly identify the dates when there are no breaks, relies on these combined results. This leads us to Theorem 2.3.

Theorem 2.3. *Suppose that Assumptions 2.1-1.4 hold and $N_g/N \rightarrow \pi_g \in (0,1)$ for any $g \in \mathcal{G}$. Then, as $N, T \rightarrow \infty$ and $N/T^\delta \rightarrow 0$ for some $\delta > 0$, we have that*

$$\mathbb{P} \left(\left\{ \left| \hat{\beta}_{k,g,t} - \hat{\beta}_{k,g,t-1} \right| = 0 \forall t \in \mathfrak{T}_{g,m_{k,g}^0}^{0c} \text{ and } g \in \mathcal{G}, k \in \mathcal{P} \right\} \right) \rightarrow 1,$$

where $\mathfrak{T}_{g,m_{k,g}^0}^{0c} := \{2, \dots, T\} \setminus \mathfrak{T}_{g,m_{k,g}^0}^0$ for all k, g and t .

This result is similar to that of Theorem 1.2, but with the difference in that we assume $N/T^\delta \rightarrow 0$. This condition is satisfied as long as N is of the geometric order of T . Moreover, Corollary 2.1 demonstrates that the procedure consistently identifies the number of breaks and the corresponding dates.

Corollary 2.1. *Suppose that Assumptions 2.1-2.4 hold. Also that $N_g/N \rightarrow \pi_g \in (0,1)$ for any $g \in \mathcal{G}$. Then for all k and g as $N, T \rightarrow \infty$ and $N/T^\delta \rightarrow 0$ for some $\delta > 0$, we have that*

1. $\mathbb{P} \left(\left\{ \hat{m}_{k,g} = m_{k,g}^0 \forall k, g \right\} \right) \rightarrow 1;$
2. $\mathbb{P} \left(\left\{ \hat{\mathfrak{T}}_{g,\hat{m}_{k,g}} = \mathfrak{T}_{g,m_{k,g}^0}^0 \forall k, g \mid \hat{m}_{k,g} = m_{k,g}^0 \forall k, g \right\} \right) \rightarrow 1.$

Corollary 2.1 implies that our approach can identify the true model with the correct number of breaks and dates, so that $\hat{\mathfrak{t}}_{k,g,j} = \mathfrak{t}_{k,g,j}^0$ for all k, g , and j with probability approaching one. The following theorem provides the asymptotic distribution of the estimator.

Theorem 2.4. Suppose that Assumptions 2.1-2.4 hold. Also, suppose that $N_g/N \rightarrow \pi_g \in (0, 1)$ for any $g \in \mathcal{G}$ and that $NT^{-\delta} \rightarrow 0$ for some $\delta > 0$. Then, conditional on $m_{k,g}^\dagger = m_{k,g}^0$ and $\dot{m}_{k,g} = \dot{m}_{k,g}^0$ for all $g \in \mathcal{G}$ and $k = 1, \dots, p$, and a fixed integer $\ell \in [1, \dot{m}^0]$, we have as $N, T \rightarrow \infty$ that

$$\mathbf{S}\sqrt{N}\mathbf{O}_0^{1/2}(\hat{\mathbf{a}} - \mathbf{a}^0) \rightarrow_d \mathcal{N}\left(\mathbf{0}_{\ell \times 1}, \mathbf{S}\mathbf{\Psi}_0^{-1}\mathbf{\Pi}^{-1/2}\mathbf{\Lambda}_0\mathbf{\Pi}^{-1/2}\mathbf{\Psi}_0^{-1}\mathbf{S}'\right).$$

Similar to the homogeneous case, the estimator is shown to be oracle efficient. Additionally, because the number of breaks is allowed to grow with the sample size ($T \rightarrow \infty$) in the heterogeneous version of the estimator, we use a selection matrix that enables us to work with finite dimensions as the number of breaks, $\dot{m}$, can diverge. Essentially, the theorem above demonstrates that, for any arbitrary linear combination of elements of $\hat{\mathbf{a}}$, the estimator for coefficient k in group g under regime j is $\sqrt{N_g I_{k,g,j}}$ consistent. To be more precise, Theorem 2.4 demonstrates that the variance of the post-selection estimator for a given group depends on the number of members in that group.

3.4.1 Choosing the tuning parameter λ_2

The estimator discussed in the previous sections is conditional on the tuning parameters λ_2 and G . Additionally, the estimator depends on the initial estimator of the group memberships, $\hat{\gamma}$. We direct you to Appendix E, which contains Algorithms 2 and 3, demonstrating how to derive the heterogeneous estimator in practice (refer also to Bonhomme and Manresa 2015; and Okui and Wang 2021). The information criterion that we minimize is given as:

$$\text{IC}_2(\lambda_2) := \sigma_{\lambda_2}^2\left(\hat{\mathbb{T}}(\lambda_2)\right) + \varphi_2 \cdot \left[\sum_{k=1}^p \sum_{g=1}^G (\dot{m}_{k,g}(\lambda_2) + 1) \right], \quad (3.2)$$

where $\hat{\mathbb{T}}(\lambda_2)$ and $\dot{m}_{k,g}(\lambda_2)$ are $\hat{\mathbb{T}}$ and $\dot{m}_{k,g}$, respectively, viewed as functions of λ_2 with $\varphi_2 = \varphi_2(N, T) > 0$ being a penalty. If we fix any $k \in \mathcal{P}$, we can define $\sigma_{\lambda_2}^2(\hat{\mathbb{T}}(\lambda_2))$ as:

$$\frac{1}{c_{NT}} \sum_{i=1}^N \sum_{j=1}^{\dot{m}_{k,g_i}(\lambda_2)+1} \sum_{t=\hat{\mathbb{T}}_{k,g_i,j-1}(\lambda_2)}^{\hat{\mathbb{T}}_{k,g_i,j}(\lambda_2)-1} \left(\tilde{y}_{it} - x_{k,it} \hat{\mathbf{a}}_{k,g_i,j} + x_{k,i1} \hat{\mathbf{a}}_{k,g_i,1} - \sum_{\ell \neq k} [x_{\ell,it} \hat{\phi}_{\ell,g_i,t} - x_{\ell,i1} \hat{\phi}_{\ell,g_i,1}] \right)^2,$$

where $c_{NT} := N(T-1)$ and $\hat{\phi}_{\ell,g_i,t} := \sum_{j'=1}^{\dot{m}_{\ell,g_i}+1} \mathbb{1}(t \in \hat{\mathbb{T}}_{\ell,g_i,j'}) \hat{\mathbf{a}}_{\ell,g_i,j'}$ is a coefficient that assigns the correct post-selection estimator for each $\ell \neq k$ variable. Similar to the approach in the previous section, we suggest adopting $\varphi_2(N, T) = c[\log(NT)/\sqrt{NT}]$ with $c = 0.05$ as typically done in the literature (see, for example, Okui and Wang 2021). In Theorem 2.5, found in Appendix D, we demonstrate that minimizing IC_2 effectively leads to the selection of an appropriate λ_2 , thereby determining the correct number of breaks for all coefficients.

4 Monte Carlo

4.1 A homogeneous simulation

We begin with an example of a panel model, represented as $y_{it} = \xi_i + \mathbf{x}_{it}'\boldsymbol{\beta}_t + u_{it}$, where the regressors $\mathbf{x}_{it}$ are generated as $\mathbf{x}_{it} = 0.2 \cdot (\mathbf{1}_{p \times 1} \cdot \xi_i) + \mathbf{e}_{it}$. In this model, both ξ_i and $\mathbf{e}_{it}$ are independent and identically distributed (i.i.d.), with ξ_i being a scalar and $\mathbf{e}_{it}$ a $p \times 1$ vector, both following a standard normal distribution for $i = 1, \dots, N, t = 1, \dots, T$, where p denotes the number of regressors. This design is similar to the one used by Wang et al. (2018). Moreover, following Bai and Ng (2002), we allow for not only serial and cross-sectional correlation but also heteroskedasticity in u_{it} such that

$$u_{it} = a_1 u_{it-1} + v_{it} + \sum_{j=1}^K a_2 (v_{i-j,t} + v_{i+j,t}), \quad (4.1)$$

where $u_{1,0}, \dots, u_{N,0} = 0$, $a_1 = 0.5$, $a_2 = 0.1$, $K = 5$ and $v_{it} \sim \mathcal{N}(0, \sigma_i^2)$ with $\sigma_i \sim U(0.5, 1)$. Here, a_1 and a_2 are the serial and cross-section correlation terms, respectively. The heteroskedasticity is allowed by the fact that v_{it} has a variance that is different across different cross-sections.

Let $\boldsymbol{\beta}_t = [\beta_{1,t}, \dots, \beta_{p,t}]'$ have component-wise breaks, where the true slope vectors, $\boldsymbol{\beta}_t^0$ is arranged in a $p \times T$ matrix with $T = 5$, $p = 6$, and $N \in \{100, 150, 200, 250, 300\}$ such that⁹

$$\mathbf{B}^0 = \begin{bmatrix} \boldsymbol{\beta}_1^{0'} \\ \boldsymbol{\beta}_2^{0'} \\ \vdots \\ \boldsymbol{\beta}_6^{0'} \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0.9 & 0.2 & 0.2 \\ 1.25 & 1.25 & -1.25 & 0.15 & 2.25 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ -0.5 & -0.5 & 0.5 & 0.5 & 0.5 \end{bmatrix}. \quad (4.2)$$

In (4.2), each column corresponds to a time point, and each row corresponds to a coefficient. Here, we allow the first coefficient to break in the third and fourth periods, the second coefficient to break in the third, fourth, and fifth periods, and the sixth coefficient to break in the third period.¹⁰ This gives us the following vector of the true number of breaks, $[m_1^0, \dots, m_6^0] = [2, 3, 0, 0, 0, 1]$. In Figure 1, we run Monte Carlo simulations to showcase the procedure suggested in Section 2.1.

⁹We set N to a relatively big number because the approach is proven to be consistent when N tends to infinite.

¹⁰The choice of the simulated $\mathbf{B}$ is not very important here. In unreported MC simulations, we tried many specifications that gave us similar conclusions. For example, we get similar results if we set $p = 4$ keeping rows 1, 2, 6 and any of the rows in between.

Before delving into the results, we note that we select the tuning parameter by minimizing (3.1). To do this, we split the tuning parameter sample into a 50 log-spaced grid in the interval $[\lambda_{\min}, \lambda_{\max}]$. Here, $\lambda_{\max}$ is a tuning parameter value that gives zero number of breaks, and $\lambda_{\min}$ gives many. We follow the convention of setting $\kappa_1 = 2$ and $c = 0.05$, as done in work such as that of Qian and Su (2016). Note that the choice of κ_1 is not of big importance as the value of the tuning parameter changes to accommodate for different values of κ_1 .

In Figure 1(a), we demonstrate the method by running it once with $N = 150$, illustrating how the method provides estimates of $\{m_k\}$ depicted by the colored lines. We observe that the minimum of the IC function (blue line) identifies the optimal value of λ_1 , which, in turn, indicates the break number for all coefficients, as shown by the vertical black dashed line. We find that $[\hat{m}_1, \dots, \hat{m}_6] = [2, 3, 0, 0, 0, 1]$, as expected. In Figure 1(b), with $N = 150$ and running the procedure $N_{\text{sim}} = 1000$ times, we derive the distributions of the breaks, where we see that $\hat{m}_k$ is mostly being estimated to the true number of breaks for all coefficients.

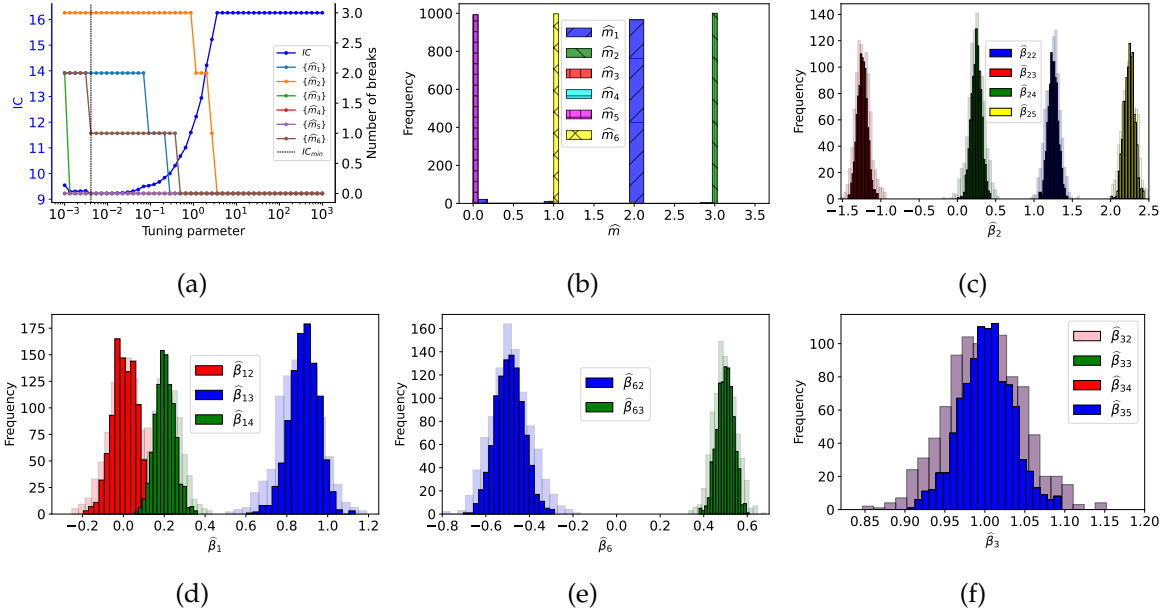


Figure 1: Plot (a) corresponds to a one replication of the CBCL break estimation procedure where we select the optimal tuning parameter by minimizing the information criterion represented by the left vertical line. Plot (b) corresponds to 1000 replications of the procedure where we plot the distribution of the breaks for each coefficient. Plots (c)-(e) correspond to 1000 replications where we plot coefficients 1, 2 and 6 pre and post-breaks for $N = 150$ and $N = 300$ depicted by the shaded and clear colors, respectively. Plot (f) is the same as (c)-(e) but we plot coefficient three for all periods.

Figures 1(c)-1(f) display histograms from 1000 replications of the coefficient estimates.

The distribution of estimated coefficients is shown for sample sizes $N = 150$ and $N = 300$, with faded and clear histograms, respectively. We see that the estimated coefficients are close to the true values. For example, the distributions of $\hat{\beta}_{1,2}$ and $\hat{\beta}_{1,3}$ center around their true values, $\beta_{1,2}^0 = 0$ and $\beta_{1,3}^0 = 0.9$. These distributions narrow as the sample size increases, indicating consistency. Another important property of the estimator can be observed in Figure 1(f). Note how $\hat{\beta}_{3,t}$ is numerically identical for all t across all simulations, implying that the method can effectively shrink non-breaking coefficients without a numerical error.

In Table 1, we compare the mean squared error (MSE) and the bias between the post-CBCL estimator and the shrinkage post-selection estimator suggested by Qian and Su (2016) for $T \in \{5, 7\}$ and N as previously set.¹¹ As expected, the MSE and the bias are lower when estimating the parameters while accounting for CBC breaks. The reason for this result is that CBC breaks provide information on which parameters are shifting, protecting against the unnecessary loss of data points.¹²

INSERT TABLE 1 ABOUT HERE

4.2 A heterogeneous simulation

In this section, we explore the heterogeneous case. Consider the panel model $y_{it} = \xi_i + \zeta_{g,t} + \mathbf{x}_{it}'\boldsymbol{\beta}_{g,t} + u_{it}$ with ξ_i , $\mathbf{x}_{it}$, and u_{it} being generated as in the previous section, and $\zeta_{g,t}$ is an intercept. Let $\boldsymbol{\beta}_{g,t}$ change component-wise across different groups. We stack the true slope vectors (with an intercept) in a $G \times (p+1) \times T$ tensor, $\mathcal{B}$. We set $T \in \{5, 10, 15\}$, $N \in \{200, 300, 400, 450\}$, $p = 5$, and $G = 3$. When $T = 5$, we let $\mathcal{B}$ contain:

$$\mathbf{B}_1 = \begin{bmatrix} 0.1 & 0.1 & 0.1 & 1 & 0.1 \\ 1.25 & 1.25 & -1.25 & 0.15 & 2.25 \\ 1 & 1 & 0 & -1 & -1 \\ 1 & 1 & 1 & 1 & 1 \\ -0.5 & -0.5 & 0.5 & 0.5 & 0.5 \\ 2 & 2 & 2 & 3 & 3 \end{bmatrix}, \quad \mathbf{B}_2 = \begin{bmatrix} 0.1 & 0.1 & 0.1 & 1 & 0.1 \\ 2 & 2 & 2 & 3 & 3 \\ 2 & 2 & 2 & 3 & 3 \\ 2 & 2 & 2 & 3 & 3 \\ 2 & 2 & 2 & 3 & 3 \\ 2 & 2 & 2 & 3 & 3 \end{bmatrix}$$

and $\mathbf{B}_3 = \mathbf{0}_{(p+1) \times T}$. When $T \in \{10, 15\}$, we set the last element of each row to match the remaining elements of the matrix. For example, in the first row of $\mathbf{B}_1$, it is represented by

¹¹Previous methods penalize the Frobenius norm difference $\|\boldsymbol{\beta}_t - \boldsymbol{\beta}_{t-1}\|$, i.e. a group fused lasso type penalty. This penalty induces vector breaks and hence one break number and one set of break dates are selected when the norm difference is greater than zero.

¹²When we increase T from $T = 5$ to just $T = 7$, we observe that our method's outperformance becomes much wider. This is intuitive because CBC breaks utilize the extra points in the time dimension.

$(0.1, 0.1, 0.1, 1, 0.1, 0.1, \dots, 0.1)$. Each element corresponds to a group, where $\mathbf{B}_g$ represents the coefficients of group g for all $g = 1, 2, 3$. Additionally, we set $N_1 = N_2 = 0.3N$ and $N_3 = 0.4N$, where N_g is the number of cross-sections in group g .

Considering $\mathbf{B}_1$, the first row represents the first coefficient of group one, set to 0.1 until the fourth period, then becoming 1 and returning to 0.1. The second coefficient breaks in the third, fourth, and fifth periods, the third coefficient in the third and fourth, and the fifth in the third period. The last coefficient is an intercept that breaks at the fourth period. For $\mathbf{B}_2$, the first and last coefficients are the same as those in the first group to allow some homogeneity across groups. The remaining coefficients in $\mathbf{B}_2$ signify vector breaks, shifting from 2 to 3. Finally, $\mathbf{B}_3$ does not break and is a matrix of zeros. This provides us with the following true break numbers: $[m_{1,1}^0, \dots, m_{6,1}^0] = [2, 3, 2, 0, 1, 1]$, $[m_{1,2}^0, \dots, m_{6,2}^0] = [2, 1, 1, 1, 1, 1]$, and $[m_{1,3}^0, \dots, m_{6,3}^0] = [0, \dots, 0]$.

Before presenting the results, we minimize the information criterion (E.5) in Appendix E to estimate the number of groups. Setting $N_{\text{sim}} = 1000$ replications, we observe that the method predominantly estimates $\hat{G} = 3$, aligning with the desired outcome. To assess the accuracy of cross-section classification into true clusters, we use the misclassification frequency (MF) measure, defined as $\text{MF} := N^{-1} \sum_{i=1}^N \mathbb{1}(\hat{g}_i \neq g_i^0)$, where $\hat{g}_i$ and g_i^0 represent the estimated and true group memberships, respectively. The Monte Carlo average of MF across each replication is around zero, indicating nearly perfect clustering accuracy for our specifications. Even when underestimating the true number of groups ($G^0 = 3$) and choosing $G = 2$, the algorithm effectively assigns cross-sections to a mix of the first two groups without contaminating the third. This ensures interpretable results even with an incorrect number of groups. Conversely, selecting too many groups ($G = 4$) results in splitting the third group into two groups, suggesting the algorithm only refines one group unnecessarily, a minor issue that diminishes as the sample size grows.

In Tables 2-5, we present the main results, including four measures of accuracy: the mean squared error (MSE), the average number of estimated breaks, the frequency of false detection of the estimated number of breaks, and the frequency of false detection of the estimated break periods given that the number of breaks is selected correctly.

INSERT TABLES 2-5 ABOUT HERE

The tables are divided into three panels for groups $g = 1, 2$, and 3, showing results

as vectors for each coefficient. Focusing on group $g = 1$ with a sample size of $N = 450$ and a time series length of $T = 5$, we observe the frequency of false detection for the estimated number of breaks as 0.009, 0.000, 0.012, 0.006, 0.018, and 0.011 for variables 1 to 6, respectively. Similarly, the simulation's average estimated number of breaks for this group is $[2.000, 3.000, 2.006, 0.016, 1.014, 1.007]$. Given the true number of breaks $[m_{1,1}^0, \dots, m_{6,1}^0] = [2, 3, 2, 0, 1, 1]$, the procedure shows good performance for this sample size. The procedure generally performs well across most sample sizes and shows improvements as the sample increases. As an example, take $g = 2$ and recall that $[m_{1,2}^0, \dots, m_{6,2}^0] = [2, 1, 1, 1, 1, 1]$. The Monte Carlo average in this case improves from $[1.965, 1.174, 1.160, 1.158, 1.182, 1.174]$ at $N = 200$ and $T = 5$, to $[2.005, 1.038, 1.050, 1.063, 1.064, 1.063]$ when $N = 450$. This consistent trend across all accuracy measures for all groups confirms the method's effectiveness even at realistic sample sizes. Overall, the results provide encouraging evidence for the method's accuracy and performance, with noted improvements as the sample size increases.

5 Empirical Illustration

5.1 Motivation

In this section, we revisit the empirical topic concerning the effects of socioeconomic determinants on crime. When studying this question, we want to look at whether one mistakenly concludes the existence of breaks in some coefficients even if they are induced by the controls. This could lead to loss of power leading to changes in the magnitude of the coefficients and their significance levels. Another point we look into is about overestimating the number of breaks because different coefficients are breaking in different periods.

When studying this, the literature tends to ignore the possibility that structural breaks could be contaminating the results. One exception is the study by Kaddoura and Westerlund (2023), which examined this question while allowing for structural breaks in the parameter vector. They focus on deterrence variables as the main drivers of crime while controlling for variables typically used in the literature (see Cornwell and Trumbull 1994, for an overview). The total variables they include is $p = 16$ which might seem reasonable at first but makes it hard to interpret what is driving the breaks. This shows in their results, where they initially find $\hat{m} = 2$ vector breaks. They also run a two-stage procedure where they start by projecting out the controls, saving the residuals, and then running their method on the residuals. They

observed no breaks after doing that which implies that some of the controls were driving their results. However, the fact that they chose which variables get to be projected out in the first stage is arbitrary and unreliable.

The previous discussion shows the need to properly handle the possibility that each coefficient has its own set of breaks. Hence, we use the CBCL break estimator.

5.2 Model

The data we use is the same one used by Cornwell and Trumbull (1994) and Kaddoura and Westerlund (2023). Because the data set has $N = 90$ and a relatively small $T = 7$, we use the estimator given in section 2.1. Hence, the model we consider is

$$\text{crime}_{it} = \sum_{k=1}^5 \beta_{k,t} D_{k,it} + \sum_{\ell=1}^{11} \theta_{\ell,t} C_{\ell,it} + \xi_i + e_{it}, \quad (5.1)$$

where crime_{it} is the crime rate and $D_{k,it}$ are deterrence variables that are usually used in the literature (see e.g., Dills et al. 2010). Mainly, we include the probability of arrest (PRBARR), the probability of conviction given arrest (PRBCONV), the probability of a prison sentence given a conviction (PRBPRIS), the average prison sentence in days (AVGSEN), and the number of police per capita (POLPC). Moreover, $C_{\ell,it}$ are the controls which include the average weekly wages in construction (WCON), transportation, utilities and communication (WTUC), wholesale and retail trade (WTRD), finance, insurance and real estate (WFIR), services (WSER), manufacturing (WMFG), federal government (WFED), state government (WSTA) and local government (WLOC), population density (DENSITY), and percent young male (PCTYMLE) are also included, as crime tends to depend on these variables. All the variables are transformed into logs including the dependent variable.

To compare our method to that of Kaddoura and Westerlund (2023), we let e_{it} contain random interactive effects. Consistently estimating the parameters of this model requires cross-sectionally demeaning the data to eliminate any source of bias coming from the interactive effects (see Kaddoura and Westerlund 2023, for a discussion). The cross-section fixed effects term can then be handled as discussed in the previous sections.

5.3 Results

When applying the procedure, we estimate $\hat{\mathbf{m}} := [\hat{\mathbf{m}}_D \ \hat{\mathbf{m}}_C]$, where $\hat{\mathbf{m}}_D$ and $\hat{\mathbf{m}}_C$ represent the vectors of component-wise breaks for the deterrence and control variables, respectively.

We begin by discussing the deterrence variables' results reported in the top panel of Table 6. When applying the CBCL procedure to the transformed data of (5.1), we find no breaks in the deterrence coefficients, $\hat{\mathbf{m}}_D := (\hat{m}_{d1}, \dots, \hat{m}_{d5}) = (0, 0, 0, 0, 0)$, where m_{dk} refers to the break of deterrence variable k . This is an important finding because if one applies previous methods, the conclusion is that there exist two structural breaks. In other words, the conclusion that these coefficients are breaking is caused by breaks occurring in the controls' coefficients.

To compare the estimates obtained through our method with those derived when allowing only for vector breaks, we present results from the method of Kaddoura and Westerlund (2023) in the lower panel of Table 6. In general, we note significant differences in the magnitudes and significance levels of the coefficients when only vector breaks are permitted. These disparities arise because unnecessary regime switching in finite data sets results in a loss of statistical power, leading to less accurate results. For instance, in the case of PRBARR, PRB-CONV, and POLPC, we observe substantial discrepancies in the magnitude of the coefficient estimates. Even more concerning is that the coefficient for PRBPRIS becomes statistically insignificant in the regimes for 1985 and 1986-1987, a finding not supported by the literature (see Bun et al. 2020), which clearly results from data loss.

A similar issue arises for AVGSEN, where the coefficient incorrectly appears to be significant in the last regime. At first glance, this implies that the average prison sentence significantly affects crime in the last regime. However, the average prison sentence is a variable that does not vary much over time which means that splitting the sample leads to having less variation within each regime which could be the reason for this result. This is concerning because, yet again, this result introduces inaccurate significance levels. From a policy point of view, these results might indicate that increasing the prison sentence would decrease crime on average in that last regime, even though that does not seem to be the case. In contrast, when running our method, we see that the relation with AVGSEN is insignificant which is a result supported by the literature (see Bun et al. 2020). This implies that having a longer prison sentence is not an effective tool to decrease expected crime. In a way, this reinforces the notion that criminals place more emphasis on avoiding conviction rather than on the severity of punishment once convicted (see Zimring and Hawkins 1973).

As for the other results provided by running our method, they are intuitive and follow economic theory. Take for example PRBARR, PRB-CONV, and PRBPRIS, where we see that the results are statistically significant and negative. This means that the conditional expected

tation of crime is reduced when the probabilities of arrest, conviction, and prison sentence given a conviction arrest increase. This is in line with previous work (see e.g., Cornwell and Trumbull 1994; Cherry and List 2002; and Ghasemi 2017). Moreover, POLPC has a statistically significant positive effect on crime which is consistent with the results of Cornwell and Trumbull (1994), Baltagi (2006), and Baltagi and Liu (2009). We emphasize, again, that if one simply allows for vector breaks then these results would have been contaminated leading to false conclusions.¹³

INSERT TABLE 6 ABOUT HERE

Let us focus on the control variables highlighted in Table 6's top panel. WTUC and PCTYMLE both show a single break in 1986, while DENSITY has one in 1985. Unlike previous methods that might identify two breaks, this approach accurately captures one break for different variables at distinct times, 1985 and 1986, detailed in the table's lower panel. This means that the CBCL break procedure provides us with more information on which relations are breaking and at what times. Moreover, the effect of DENSITY is significant and negative but seems to be decreasing from regime 1982-1984 to regime 1985-1987. Both WTUC and PCTYMLE significantly affect expected crime, with WTUC increasing and PCTYMLE decreasing when moving from the 1982-1985 regime to the 1986-1987 regime. The remaining controls in the lower panel are insignificant, aligning with findings from previous work (see e.g., Cornwell and Trumbull 1994; Baltagi 2006; and Baltagi and Liu 2009). However, this is not the case when we only allow for vector breaks; we observe that some of these controls exhibit significant coefficients across different regimes.

Allowing for breaks is typically viewed as an extension and, in some cases, a crucial aspect of certain applications. However, as we observed, if we had not treated these breaks component-wise, we would have incorrectly concluded that the deterrence variables we focus on have two breaks. This led to inaccurate conclusions regarding the relationship between crime and these variables. In a way, allowing for vector breaks caused more harm than good in this situation, implying that one has to be careful in understanding which relationships are breaking and why. Our method protects against such concerns and contributes

¹³Cornwell and Trumbull (1994) discussed the possibility of PRBARR and POLPC being endogenous. They also use the fraction of crimes that involve face-to-face contact, and per capita tax revenue as possible instruments. However, the only significant difference observed is that the GMM's standard errors are larger. We hence only report the results given by the CBCL break estimator without using a GMM penalized version of (2.3).

to the crime literature by showcasing that what has been found until now is robust to breaks.

6 Conclusion

In this article, we address a concern that arises when estimating structural breaks, particularly when an *unknown* subset of coefficients is subject to breaks. Therefore, we consider the estimation of what we refer to as “coefficient-by-coefficient breaks” or “component-wise breaks,” allowing each component of the coefficient vector to exhibit breaks. We derive a novel estimator, the coefficient-by-coefficient Lasso (CBCL) break estimator, that utilizes coefficient-specific adaptive weights and incorporates an ℓ_1 fused penalty to induce the sparsity of component-wise breaks. Furthermore, we consider two versions of the estimator, one with homogeneous breaks and the other with heterogeneous breaks, making it suitable for various empirical scenarios. Our asymptotic results demonstrate that the CBCL estimator consistently identifies breaks, groups, and break dates for all coefficients. Additionally, the CBCL post-selection estimator is shown to be asymptotically normal. These findings are further supported by Monte Carlo simulations. The importance of component-wise breaks is also highlighted through an empirical application of crime data.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at [\[link here\]](#).

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Table 1: Simulation results for homogeneous breaks.

N	T	MSE_{CBC}	MSE_{VB}	$Bias_{CBC}$	$Bias_{VB}$
100	5	0.03162	0.03700	0.02000	0.05000
150	5	0.00775	0.00945	-0.02000	0.03798
200	5	0.00160	0.00705	-0.00959	0.03170
250	5	0.00101	0.00200	0.00856	0.01001
300	5	0.00080	0.00100	0.00625	0.01000
100	7	0.00741	0.02901	-0.02000	-0.05000
150	7	0.00255	0.00821	0.00146	-0.01773
200	7	0.00130	0.00641	-0.00200	0.00500
250	7	0.00027	0.00100	-0.00200	0.00600
300	7	0.00020	0.00241	0.00600	0.00700

Notes: The terms “ MSE_{CBC} ” and “ $Bias_{CBC}$ ” represent the MSE and Bias for the homogeneous CBCL post-selection estimator, while “ MSE_{VB} ” and “ $Bias_{VB}$ ” correspond to the vector breaks shrinkage estimator of Qian and Su (2016). The values reported in the table are the average MSE and average bias values across $N_{sim} = 1000$ replications, each multiplied by 100.

Table 2: Simulation Results: Average MSE Values for Heterogeneous Breaks

T	N	$g = 1$	$g = 2$	$g = 3$
5	200	0.00321	0.00360	0.00003
5	300	0.00240	0.00311	0.00003
5	400	0.00213	0.00282	0.00002
5	450	0.00201	0.00250	0.00002
10	200	0.00301	0.00316	0.00000
10	300	0.00240	0.00249	0.00000
10	400	0.00209	0.00226	0.00000
10	450	0.00191	0.00197	0.00000
15	200	0.00300	0.00284	0.00000
15	300	0.00240	0.00241	0.00000
15	400	0.00209	0.00230	0.00000
15	450	0.00191	0.00190	0.00000

Notes: In this table, we report the mean squared error (MSE) from running $N_{sim} = 1000$ replications of the heterogeneous estimator. The MSE is calculated as $((T - 1)p)^{-1} \sum_{\ell=1}^p \sum_{t=2}^T (\hat{\beta}_{k,g,t} - \beta_{k,g,t}^0)^2$ for $g \in \{1, 2, 3\}$. Here, T represents the number of periods, and p is the number of included variables.

Table 3: Simulation results for heterogeneous breaks: Average of breaks.

T	N	$g = 1$	$g = 2$	$g = 3$
5	200	[1.950, 3.000, 2.081, 0.165, 1.146, 1.135]	[1.965, 1.174, 1.160, 1.158, 1.182, 1.174]	[0.081, 0.139, 0.122, 0.133, 0.121, 0.122]
5	300	[2.005, 3.000, 2.050, 0.080, 1.100, 1.060]	[2.011, 1.101, 1.087, 1.084, 1.131, 1.115]	[0.025, 0.055, 0.050, 0.063, 0.070, 0.060]
5	400	[2.003, 3.000, 2.022, 0.045, 1.032, 1.025]	[2.005, 1.061, 1.056, 1.056, 1.080, 1.044]	[0.017, 0.029, 0.039, 0.032, 0.024, 0.034]
5	450	[2.002, 3.000, 2.024, 0.032, 1.030, 1.028]	[2.005, 1.038, 1.050, 1.063, 1.064, 1.063]	[0.010, 0.021, 0.017, 0.028, 0.016, 0.017]
10	200	[1.960, 3.000, 2.100, 0.100, 1.110, 1.110]	[1.960, 1.140, 1.121, 1.136, 1.118, 1.122]	[0.035, 0.088, 0.109, 0.086, 0.081, 0.069]
10	300	[2.005, 3.000, 2.040, 0.055, 1.051, 1.041]	[2.005, 1.061, 1.067, 1.050, 1.070, 1.068]	[0.019, 0.040, 0.029, 0.033, 0.033, 0.044]
10	400	[2.004, 3.000, 2.018, 0.020, 1.015, 1.021]	[2.005, 1.032, 1.032, 1.030, 1.030, 1.025]	[0.006, 0.017, 0.013, 0.012, 0.013, 0.007]
10	450	[2.000, 3.000, 2.013, 0.009, 1.013, 1.012]	[2.000, 1.018, 1.021, 1.023, 1.031, 1.010]	[0.006, 0.008, 0.011, 0.011, 0.008, 0.009]
15	200	[1.900, 3.000, 2.006, 0.100, 1.110, 1.100]	[1.900, 1.122, 1.102, 1.122, 1.100, 1.100]	[0.060, 0.067, 0.066, 0.077, 0.065, 0.065]
15	300	[1.995, 3.000, 2.045, 0.050, 1.039, 1.040]	[1.999, 1.057, 1.045, 1.058, 1.050, 1.035]	[0.020, 0.029, 0.037, 0.032, 0.041, 0.034]
15	400	[2.004, 3.000, 2.017, 0.017, 1.019, 1.022]	[2.000, 1.018, 1.023, 1.033, 1.020, 1.027]	[0.008, 0.009, 0.008, 0.014, 0.006, 0.012]
15	450	[2.000, 3.000, 2.006, 0.016, 1.014, 1.007]	[2.000, 1.013, 1.011, 1.016, 1.010, 1.017]	[0.006, 0.006, 0.004, 0.010, 0.005, 0.006]

Notes: In this table, we report the average of estimated breaks given by $[\bar{m}_{1,g}, \dots, \bar{m}_{6,g}]$, where $\bar{m}_{k,g} := 1/N_{\text{sim}} \sum_{j=1}^{N_{\text{sim}}} \hat{m}_{k,g,j}$ with $\hat{m}_{k,g,j}$ being the estimated number of breaks for coefficient k in group g at replication j . The number of replications we set is $N_{\text{sim}} = 1000$.

Table 4: Simulation results for heterogeneous breaks: Incorrect estimation of breaks.

T	N	$g = 1$	$g = 2$	$g = 3$
5	200	[0.038, 0.000, 0.105, 0.100, 0.140, 0.130]	[0.072, 0.167, 0.149, 0.143, 0.168, 0.162]	[0.081, 0.121, 0.106, 0.122, 0.106, 0.108]
5	300	[0.007, 0.000, 0.045, 0.092, 0.094, 0.056]	[0.022, 0.097, 0.084, 0.082, 0.125, 0.110]	[0.025, 0.049, 0.043, 0.060, 0.060, 0.052]
5	400	[0.003, 0.000, 0.025, 0.035, 0.033, 0.025]	[0.005, 0.056, 0.053, 0.055, 0.078, 0.044]	[0.017, 0.026, 0.036, 0.031, 0.022, 0.032]
5	450	[0.002, 0.000, 0.024, 0.003, 0.028, 0.027]	[0.005, 0.038, 0.047, 0.063, 0.064, 0.060]	[0.011, 0.020, 0.017, 0.026, 0.015, 0.016]
10	200	[0.084, 0.000, 0.125, 0.076, 0.105, 0.108]	[0.090, 0.122, 0.107, 0.123, 0.109, 0.111]	[0.034, 0.084, 0.102, 0.077, 0.075, 0.065]
10	300	[0.019, 0.000, 0.041, 0.028, 0.051, 0.039]	[0.022, 0.057, 0.057, 0.054, 0.065, 0.063]	[0.019, 0.039, 0.028, 0.031, 0.033, 0.039]
10	400	[0.004, 0.000, 0.018, 0.012, 0.015, 0.012]	[0.010, 0.031, 0.029, 0.025, 0.028, 0.022]	[0.006, 0.016, 0.013, 0.012, 0.013, 0.007]
10	450	[0.006, 0.000, 0.013, 0.006, 0.013, 0.011]	[0.007, 0.017, 0.020, 0.022, 0.030, 0.010]	[0.006, 0.007, 0.010, 0.010, 0.008, 0.009]
15	200	[0.198, 0.000, 0.134, 0.098, 0.119, 0.097]	[0.211, 0.111, 0.095, 0.115, 0.098, 0.093]	[0.060, 0.063, 0.064, 0.074, 0.064, 0.078]
15	300	[0.044, 0.000, 0.047, 0.050, 0.038, 0.042]	[0.042, 0.054, 0.042, 0.056, 0.050, 0.033]	[0.020, 0.028, 0.037, 0.032, 0.039, 0.034]
15	400	[0.010, 0.000, 0.017, 0.011, 0.019, 0.022]	[0.012, 0.016, 0.022, 0.029, 0.020, 0.024]	[0.008, 0.009, 0.008, 0.014, 0.006, 0.012]
15	450	[0.009, 0.000, 0.012, 0.006, 0.018, 0.011]	[0.009, 0.024, 0.015, 0.019, 0.014, 0.020]	[0.006, 0.006, 0.004, 0.010, 0.005, 0.005]

Notes: In this table, we report the average of incorrectly estimating the number of breaks given by $[IEB_{1,g}, \dots, IEB_{6,g}]$, where $IEB_{k,g} := 1/N_{\text{sim}} \sum_{j=1}^{N_{\text{sim}}} IEB_{k,g,j}$ with $IEB_{k,g,j} := \mathbb{1}(\hat{m}_{k,g,j} \neq m_{k,g}^0)$ being an indicator equal to one when the break estimate is misclassified for coefficient k , group g at replication j . The number of replications we set is $N_{\text{sim}} = 1000$.

Table 5: Simulation results for heterogeneous breaks: incorrect estimation of break periods.

T	N	$g = 1$	$g = 2$	$g = 3$
5	200	[0.001, 0.000, 0.006, 0.000, 0.000, 0.000]	[0.001, 0.011, 0.008, 0.008, 0.006, 0.006]	[0.000, 0.000, 0.000, 0.000, 0.000, 0.000]
5	300	[0.000, 0.000, 0.001, 0.000, 0.000, 0.000]	[0.000, 0.002, 0.000, 0.001, 0.001, 0.000]	[0.000, 0.000, 0.000, 0.000, 0.000, 0.000]
5	400	[0.000, 0.000, 0.000, 0.000, 0.000, 0.000]	[0.000, 0.000, 0.000, 0.001, 0.000, 0.000]	[0.000, 0.000, 0.000, 0.000, 0.000, 0.000]
5	450	[0.000, 0.000, 0.000, 0.000, 0.000, 0.000]	[0.000, 0.000, 0.000, 0.000, 0.000, 0.001]	[0.000, 0.000, 0.000, 0.000, 0.000, 0.000]
10	200	[0.002, 0.000, 0.007, 0.000, 0.000, 0.000]	[0.000, 0.003, 0.000, 0.004, 0.004, 0.001]	[0.000, 0.000, 0.000, 0.000, 0.000, 0.000]
10	300	[0.000, 0.000, 0.000, 0.000, 0.000, 0.000]	[0.000, 0.000, 0.000, 0.000, 0.000, 0.000]	[0.000, 0.000, 0.000, 0.000, 0.000, 0.000]
10	400	[0.000, 0.000, 0.000, 0.000, 0.000, 0.000]	[0.000, 0.000, 0.000, 0.000, 0.000, 0.000]	[0.000, 0.000, 0.000, 0.000, 0.000, 0.000]
10	450	[0.000, 0.000, 0.000, 0.000, 0.000, 0.000]	[0.000, 0.000, 0.000, 0.000, 0.000, 0.000]	[0.000, 0.000, 0.000, 0.000, 0.000, 0.000]
15	200	[0.002, 0.000, 0.005, 0.001, 0.001, 0.000]	[0.008, 0.003, 0.002, 0.003, 0.001, 0.000]	[0.000, 0.000, 0.000, 0.000, 0.000, 0.000]
15	300	[0.000, 0.000, 0.000, 0.000, 0.000, 0.000]	[0.000, 0.000, 0.000, 0.000, 0.000, 0.000]	[0.000, 0.000, 0.000, 0.000, 0.000, 0.000]
15	400	[0.000, 0.000, 0.000, 0.000, 0.000, 0.000]	[0.000, 0.000, 0.000, 0.000, 0.000, 0.000]	[0.000, 0.000, 0.000, 0.000, 0.000, 0.000]
15	450	[0.000, 0.000, 0.000, 0.000, 0.000, 0.000]	[0.000, 0.000, 0.000, 0.000, 0.000, 0.000]	[0.000, 0.000, 0.000, 0.000, 0.000, 0.000]

Notes: In this table, we report the frequency of incorrect estimation of break periods given by $[IEP_{1,g}, \dots, IEP_{6,g}]$, where $IEP_{k,g} := 1/N_{\text{sim}} \sum_{s=1}^{N_{\text{sim}}} IEP_{k,g,s}$ with $IEP_{k,g,s} := \sum_{j=1}^{\hat{m}_{k,g}} \mathbb{1}(\hat{T}_{k,g,j,s} \neq T_{k,g,j}^0)$ representing the incorrect detection of break period of coefficient k , group g at regime j in iteration s . The number of replications we set is $N_{\text{sim}} = 1000$.

Table 6: Empirical estimation results.

Regressor	No breaks	CBC			FE
		1982–1984	1985	1986–1987	
PRBARR	−0.4186***				−0.354***
PRBCONV	−0.2944***				−0.282***
PRBPRIS	−0.1813***				−0.173***
AVGSEN	−0.0130				−0.003
POLPC	0.4676***				0.413**
DENSITY		−1.7030*** ↗	−0.6722***	−0.6722***	0.414
WCON	−0.0161				−0.038
WTUC		0.0404*	0.0404* ↗	0.3600**	0.046***
WTRD	0.0583				−0.021
WFIR	−0.0182				−0.004
WSER	0.0099				0.009
WMGF	−0.0639				−0.360***
WFED	−0.3149				−0.309*
WSTA	−0.0304				0.053
WLOC	0.1239				0.182*
PCTYMLE		2.3136***	2.3136*** ↗	1.220**	0.627*

Regressor	VB		
	1982–1984	1985	1986–1987
PRBARR	−0.459***	−0.613***	−0.301***
PRBCONV	−0.271***	−0.422***	−0.222***
PRBPRIS	−0.196***	−0.196	−0.128
AVGSEN	0.057	−0.084	−0.134**
POLPC	0.483***	0.548***	0.414***
DENSITY	−1.7049***	−1.9894***	−0.2714**
WCON	0.017	0.490**	0.358**
WTUC	0.026	0.778*	0.084
WTRD	0.040	−0.103	−0.038
WFIR	−0.002	0.080	0.025
WSER	−0.182	0.548	−0.217
WMGF	−0.047	−0.080***	−0.080
WFED	−0.120	−0.124	0.023
WSTA	−0.030	−0.738	0.239
WLOC	−0.005	−0.149	−0.293
PCTYMLE	2.504***	1.608	1.217**

Notes: The top panel referred to as “CBC” refers to the estimates of Coefficient-by-Coefficient breaks given by our method, while the lower panel refers to vector break estimates given by the method of Kaddoura and Westerlund (2023). The dependent variable is the crime rate (CRMTE). The deterrence regressors are cases the probability of arrest (PRBARR), the probability of conviction given arrest (PRBCONV), and the probability of a prison sentence given a conviction (PRBPRIS). The included control variables are average prison sentence in days (AVGSEN), the number of police per capita (POLPC), population density (DENSITY), percent young male (PCTYMLE), and the average weekly wage by industry, where the included industries are construction (WCON), transportation, utilities and communication (WTUC), wholesale and retail trade (WTRD), finance, insurance and real estate (WFIR), services (WSER), manufacturing (WMFG), federal government (WFED), state government (WSTA), and local government (WLOC). All variables are expressed in logs. The column labeled “No breaks” contains the variables corresponding to the coefficients without breaks. The rest of the columns represent the different regimes and ↗ indicates when a variable’s coefficient is shifting from one regime to another. If there is no arrow, it means the regime extends to the years after. In the VB panel, all coefficients are shifting. The final column represents the two-way fixed effects estimator. The symbols *, ** and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.